

ON THE PROBLEM OF T ACTIONS

[919]

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1. The problem is: *Given three points and a circle (the points and the circle being in the same plane), to find a circle through the three points and touching the given circle.* There are obviously seven distinct cases, viz. this is so when

Case 1. The three points are in a circle, that is, lie in a circle which does not meet the given circle, or in a circle which meets it in two points but we have to do with the circle through them and the given circle, or, lastly, in a circle which touches the given circle.

Case 2. The second, third, and fourth problems are, it is clear, derived from the first problem by the change of $(3, c)$ into $(-3, c)$, $(3, -c)$, $(-3, -c)$ respectively; so that only the first problem need be considered, viz. this is as shown

$$r = a + 3, \quad s = b + 3, \quad t = c + 3,$$

where also

$$r - 1 = b - c, \quad t - r = a - c, \quad t - s = a - b,$$

where $b - c$, $c - a$, $a - b$ are given magnitudes the algebraical sum of which is $= 0$, viz. they are of these positive, and the remaining one negative.

2. The most simple and straightforward geometrical solution is that given in the *Principia*, Book I, Lemma XVI. I reproduce this as given in Motte's translation (*The Mathematical Principles of Natural Philosophy*, by Sir Isaac Newton, translated into English by Andrew Motte, 8^{vo}., 2 vols., London, 1729):

"Lemma XVI. *From three given points to draw to a fourth point which is not given three right lines whose differences shall be either given or none at all.*

Case 1. Let the given points be A, B, C (see figure) and the fourth point which we are to find: because of the given difference of the lines AE, BE , the locus of the point E will be a hyperbola whose foci are A, B and the principal axis is the given difference. Let that axis be MN . Taking PM to MA as MN is to AB , erect PR perpendicular to AB , and let fall ER perpendicular to PR ; then from the nature of the hyperbola ER will be to AE as MN is to AB . And by the like argument the locus of the point E will be another hyperbola, whose foci are A, C , and whose principal axis is the difference between AE and CE ; and QS a perpendicular on AC may be drawn to which QS , if from any point E of this hyperbola a perpendicular ES is let fall, this (ES) shall be to AE as the difference between AE and CE is to AC . Wherefore the ratios of ER and ES to AE are given and consequently the ratio of ER to ES one to the other: and therefore if the right lines RP, QS meet in T , and TZ and TA are drawn, the figure $TRES$ will be given in species, and the right line TE , in which the point E is somewhere placed, will be given in position. There will be given also the right line TA , and the angle ATE ; and because the ratios of AE and TE to ES are given, their ratio to each other is given also; and thence will be given the triangle ATE whose vertex is the point E . Q.E.I."

Case 2. If two of the three lines, for instance AZ and BZ , are equal, &c.

Case 3. If all the three are equal, &c.

This problematic lemma is likewise resolved in Apollonius's Book of Tactions restored by Vieta.

3. Newton, in fact, considers the hyperbolas AB and AC , each of given axis, having the foci (A, B) and (A, C) respectively, and having PR, QS for the directrices which in the two hyperbolas respectively belong to the common focus A . The required point Z thus lies on a given line through the intersection T of these two directrices; and its position on this line is determined by the condition that the distances AZ, TZ shall be in a given ratio: the locus of

the points which satisfy this last condition is of course a circle; and the position of Z is thus determined as the intersection of the given line by a given circle 1 will call a Newton-circle); there are two intersections giving points Z, Z_1 which are the centres of the two tangent circles respectively: and the line as a focus is one of these two points is of course a determinate line, but Newton's circle is only one of a singly infinite series of circles through the two points: any other solution of the problem gives therefore the same line, but not in general the same circle.

4. In what immediately follows, I use for convenience the letter F in place of the foregoing letter T .

Effecting Newton's construction, first as above, with the points $A(B, C)$; and then in like manner, secondly with the points $B(C, A)$, and thirdly with the points $C(A, B)$; and in regard to a hyperbola AB or BA , writing AB for the directrix which belongs to the focus A , and BA for the directrix which belongs to the focus B ; then

For hyperbolas AB, AC , we have intersection of directrices AB, AC is a point F ; for hyperbolas BC, BA , we have intersection of directrices BC, BA is a point G ; for hyperbolas CA, CB , we have intersection of directrices CA, CB is a point H .

Hence these three points F, G, H lie in a line, which is the line containing the required points Z_1, Z_2 ; or say it is the line Z_1Z_2 .

The points Z_1, Z_2 are determined as the intersections of this line by a circle which is the locus of the points whose distances A, F are in a given ratio; similarly they are determined as the intersections by a circle which is the locus of the points whose distances from B, G are in a given ratio; and they are determined as the intersections by a circle which is the locus of the points whose distances from C, H are in a given ratio. We have thus three Newton-circles: if the centres of these are F', G', H' respectively, then clearly these points lie on a line $F'G'H'$, which bisects at right angles the line (or chord) Z_1Z_2 ; the points A, F, F' are obviously in a line, as are also the points B, G, G' , and the points C, H, H' ; or (what is the same thing) considering for a moment the lines $F'G'H'$ as a given line bisecting Z_1Z_2 at right angles, then the centres F', G', H' would be found as the intersections of this line $F'G'H'$ with the lines AF, BG, CH respectively.

The points Z_1, Z_2 being determined as above, then the points of contact u, v_1, v_2 of the circle Z_1 with the circles A, B, C respectively are points of intersection of the three lines A, Z_1, B, Z_1, CZ_1 with these circles respectively; and similarly the points of contact u_1, v_1, w_1 of the circle Z_2 , with the same circles are points of intersection of the lines AZ_2, BZ_2, CZ_2 with these three circles respectively.

5. I compare with Newton's the construction in which the centres Z_1, Z_2 are determined by means of the points of contact with the given circles; I may refer to Prop. 92, pp. 119—120 of Casey's *Sequel to Euclid* (12^{mo}, Dublin, 1881). We have here the line Z_1Z_2 determined as the line through the radical centre O of the three circles A, B, C perpendicular to the axis of symmetry (say the axis containing the three centres of similitude of the same three circles, viz. in the figure Σ). But this last line is by Newton's construction the line $F'G'H'$, viz. this is in fact a line through O perpendicular to the line Z_1Z_2 .

6. Taking the equations of the three circles to be

$$(x - a)^2 + (y - a')^2 = a^2,$$

$$(x - b)^2 + (y - b')^2 = b^2,$$

$$(x - c)^2 + (y - c')^2 = c^2,$$

let Ξ will first find, by a separate analytical investigation, an expression for the length of the chord Z_1Z_2 .

Writing f, g, h for the distances BC, CA, AB of the three points A, B, C from each other, viz. here—see figure 4 (xy and 3 (xyz)—the square A^2, B^2 where we are no further considering the several points of the three Newton-circles A, F' , the resolution of the rectangle figure A, B, C, F'

where shown originally were A, F' , and the resolution of the same figure with A, B, C, F' , then the equations of two figures yielding the same value, are

$$\Delta x^2 + 2H \pm 2D(\xi + 5\eta + 2(2\xi + 5\eta) + 2D)\xi, 5\eta + 0,$$

or say

$$\Delta x^2 + 2H \pm 2D(x + 5y) + 2(x + 5y) = 0, \quad x = -5y.$$

And finally for the four points a, β, γ, δ of the four circles a, b, c , the rectangle yields the relation

$$\Delta a' - 2(K + K').$$

To verify this, observe that the left-hand side is

$$2(K' + K')(\alpha Z + \alpha Y) + ((3 - y)\xi - \alpha Y) + (3 - y)\xi - \alpha Y,$$

or putting hence $X + Y + Z$ for this

$$(K' + K')(\alpha + \alpha Y + \alpha Z + \alpha)Y,$$

which is then

$$\begin{aligned} &= \alpha\alpha((B - \alpha)X + (Y + (Z - \alpha) + \alpha(b - \alpha)KZY))Y + Y + Z, \\ &\quad + \alpha((\gamma - a)Z + (\alpha - \alpha)Y), \end{aligned}$$

The equation is to be verified thus becomes

$$(K' + K')((y - \alpha)X + (Y - \alpha + \cos \alpha XY).$$

This breaks up into two equations,

$$\begin{aligned} &K[(y - \alpha)X + Y + (Z + \alpha)Y] = \alpha(X - y)2Z \\ &+ \alpha\beta(\eta - \alpha)Z + (\gamma - \alpha)Y)Z; + \alpha\eta - 5Y + Z)Z; \\ &+ b(y - \beta)X - K(\alpha - \beta), -\alpha(X - \beta)X + Y + Z; \end{aligned}$$

and a like equation with the different letters. And the equation just written down, observing that each side is a linear function of X and Y , again breaks up into the two equations

$$\begin{aligned} &(\alpha - \alpha)K - 2\alpha\alpha(\beta - \gamma) - \alpha(y - \alpha) - \alpha(\alpha - \beta), \\ &\alpha(\alpha - \alpha)\beta + b - \alpha(\beta + (\gamma) - 2\alpha(\alpha - \beta), \end{aligned}$$

which are at once verified: in fact, for the first equation the right-hand side is

$$\begin{aligned} &= (\alpha - 6\alpha) + (3\alpha + 4\alpha)\alpha = \alpha + \dots \\ &= \alpha + \beta - \alpha(\beta - \alpha)\beta + b - 5, \\ &= -\alpha + \alpha(\beta - \beta), \end{aligned}$$

and similarly the second equation, the right-hand side is

$$= \alpha - b((\alpha + \beta\beta + b - \alpha)Z = (\alpha - b\beta)y + b)y.$$

Writing

$$\begin{aligned} &\Phi = K(\beta\alpha(\beta - \gamma)X - Y - Z) \\ &+ \alpha(\delta y - \alpha)Y + X + Y + Z - \beta + b\alpha(\alpha - \beta)Y \end{aligned}$$

$$+b(\alpha(\beta - \beta)Z + X - Y - Z) \\ +\alpha(\delta y - \alpha)X - Z + X + Y - \alpha((\gamma - \beta)Y + X - Y + Z),$$

we have

$$\Phi = (K, H_n + K)(1 - 2)(K - K' - 1Y)\Phi.$$

This equation determines Φ , and thus the equation of the line of centres is

$$2(K' - K)(\beta + 2)(Y + 5Y) + (KH + K, H_n) = \Phi = 0.$$

18. This line meets

$$= 2K_n + 2K_n + 8 + ax = 0,$$

in the mid-point of the chord Z_1Z_2 . We thus have for the coordinates x, y of this mid-point

$$2(K' + K')(K' + Y) + 8 + Z(Y_n) = (KH_n + K)Z - K_n(K' + K' + K_n)(\alpha + b\alpha + b\alpha)Y = 0,$$

$$2(K' + K')(K' + Y) + 5(Y) - 2((KH_n + K, H_n) - 1)(K' + K' + K_n)(b + b\alpha + b\alpha)Y = 0.$$

The perpendicular distance of the centre F' from the chord Z_1Z_2 is

$$= \frac{(\beta + b\alpha + b\alpha + \alpha\beta + \alpha\alpha)}{2(Y + K')},$$

and hence

$$2(K + K_n - \frac{1}{2(Y + K')}(2(K' + K'(Y + X) + (K_n + K' - 1))(\ell\delta + \ell\delta_n)).$$

Then

$$\frac{1}{(K - K' - 1)} = 2(K' - K'(Y + X)) + 2((K + K')(K' + K')(K - K') + (KH + K' + K' + K' + K_n)(\alpha + b\alpha + b\alpha + \alpha\beta + \alpha\alpha))$$

Thus the perpendicular distance of the centre F from the chord Z_1Z_2 is

$$= \frac{1}{2(K' + K')(K' + K')(K' - K' + 1)} = 2(KH + K' + K' + K_n) - K' - K' - K_n - K_n,$$

moreover, by what precedes, we have

$$\text{having } K = \frac{1}{2(K' + K'(K' + Y))} = -2(\alpha + \alpha\beta) - \alpha\alpha - 2\alpha(Y).$$

19. Hence also

$$(KZ_n)_n = \frac{(K' + K' - 1)}{2(K' + K' - 1)}((1 + F)(K(\gamma + Z) + (1 + F\alpha - 2\alpha))Y),$$

Supposing $A = 0$, then $\Phi - 8H = 0$, that is, the two radii are equal only if one of the factors $f - (\xi - r)$, $g + (b - r)\beta + (a - r)$ vanish, but in this case we have also $K, K' = 0$; the two tangent circles are here coincident. The equation $\zeta\delta - (a + 5)\beta$ signifies that the circle X, F touch each other from like sides intentially; that is, the curve K_n on its centre on the same line AB as in the curve K ; for these particular cases, AK is to circles centres on the same line as AB , and is the distance of the two centres is $AB - KK = 0$, which is here $b - a + a - b = 0$, that is, the same.

To show that $\Phi = 0$ is the condition of a common tangent, suppose that the three circles have a common tangent, and let P, Q, R denote the distances of the points of contact from any fixed point on this tangent; we have

$$f^2 = (Q - R)^2 + (b - c)^2 = (P - Q)^2,$$

$$g^2 = (R - P)^2 + (\alpha - c)^2 = a^2,$$

$$h^2 = (P - Q)^2 + (a - b)^2 = h^2,$$

equations which I represent by f, g, h , with $a^2 = (P - X)^2$, &c., where $A + B + Y = 0$. We have

$$\Delta = f + g^2 + h^2 + a^2 + b^2 + c^2 - 2(fg + gh + hf) - 2(\beta\gamma + \gamma\alpha + \alpha\beta)$$

and finding the corresponding values of $f^2 - g^2 + h^2$, $f^2 + g^2 - h^2$ and multiplying by $f^2 + g^2 - h^2$, &c., the equation becomes

$$= \Delta = -2a^2\beta\gamma - 2\Delta(\beta + b\gamma)F + (\gamma + Y)\Delta - \beta b 2\Delta b\alpha)b,$$

But this last three terms hence are

$$\Delta - 2(a^2\beta\gamma + 5a\beta\gamma + b\Delta), -4\Delta\Delta,$$

that is,

$$\Delta = -a/v\beta - \Delta\beta\alpha + b\beta\Delta).$$

or replacing f, a, b by their values, we have

$$\Delta - (4\beta\gamma + 4\xi) - (b\alpha + \alpha)\Delta b\gamma) - b((\alpha + \beta\beta + \alpha)\gamma),$$

which is the required condition $\Phi = 0$.

10. It would at first sight appear that the distance ZZ of the two centres would always vanish if $\Phi = 0$. But if A, B, C are real circles, this condition $\Phi = 0$ implies

$$\frac{f^2}{((b - c)^2 + a^2)} \cdot \frac{g^2}{(c - a)^2 + b^2} \cdot \frac{h^2}{(a - b)^2 + c^2},$$

where $\Delta = 0$, and the being so we have $0 = -\Delta = 0$, &c. and the value of $E\Sigma$ instead of being $= 0$, or appears to be infinite. In proof, take for a moment the origin at E , and the line $T\Sigma$ for the axis of x ; we have then $(0, b)$ for the coordinates of E . Recalling (x, b) to coordinates of E , E, E , we have $p = x + p, q = y$, for the corresponding values of p, q , and the value of $E\Sigma$ as before; we now have for the corresponding values of p, q , the values which give

$$fp = aa(A + b^2(b + b) + a^2 = aa\beta + b^2(b + b) = a(\xi\xi),$$

and thus, for the values of x, y , in conjunction with the foregoing condition $\Phi = 0$,

$$y\beta = m(B + 1)\beta - y^2 = y^2 = Dx\sigma\delta.$$

$$= -(m\delta + m^2 + y^2),$$

where A, B, C have the values $(\ell + \alpha) = a(\alpha)y, -a, A$, &c.; that is, we have in this equation the condition of $\Phi = 0$; in fact, writing $T \cdot \alpha$, &c. βb , we have the values of the centres of the circles E_n , in $(\Sigma F^2 + Z^2)$ for some values of r, y .

11. For the rectangular system: the condition $\Phi = 0$ does not imply $\Delta = 0$, and supposing $K = 0$, the distance $Z = 0$, the equation $\Phi - BH = 0$, is not satisfied, and thus the two radii are unequal; it would seem that we have concentric circles Z_n, Z_n each touching the three circles centres on A, B, C , and this would imply that the radii a, b were equal to each other; this cannot be the case, for the only solution is that given by the foregoing condition $\Phi = 0$. The explanation of this paradox is that the two circles Z_n, Z_n are not really concentric, but it is only the distance Z_n of the centres which is $= 0$, the centres are points on an imaginary line $x = m_n(g - y) = 0, (\Phi = 0)$.

In verification hereof, I start from two circles Z_n, Z_n ,

$$(x+1)^2 + (y-1)^2 = a^2,$$

$$(x-1)^2 + (y-1)^2 = a^2,$$

having for centres the two points $(-1, -1), (1, 1)$; the distance of which two points from each other is $= 2\sqrt{2}$. Consider for a moment a conic having these two imaginary points for its foci; viz. writing ξ, η for the coordinates of a point of the conic, the equation is

$$\sqrt{((\xi+1)^2 + (\eta+1)^2)} + \sqrt{((\xi-1)^2 + (\eta-1)^2)} = c^2,$$

we thence obtain

$$((\xi+1)^2 + (\eta+1)^2)((\xi-1)^2 + (\eta-1)^2) = (c^2 - (x-1)(g-1))^2,$$

that is,

$$\xi^2 - \eta^2 + 2\xi - 2\eta + 4 + 2(2(g-1) - 2(g-1)(g-1) + 2g - 2\xi\eta + (\xi+1) - 2(\xi-1)(g+1)) = (c^2 - (x-1)(g-1))^2.$$

or putting $m = u = u = 2k$, we have

$$\xi + u - 4t + b(\xi^2 - 1)^2 = (a - 4t)^2,$$

and hence

$$2 = 2(2 + 4)\xi - 4t.$$

for the equation of the conic. The last preceding equation gives

$$u(2\xi - 1)^2 + (n + a) = b - (a + ut)^2.$$

or say

$$u^2(\xi^2 + 1)^2 + (n + a) - 4 = (b + ut)^2.$$

and we have similarly

$$u^2(\xi^2 + 1)^2 + (n + a)^2 = (b + ut)^2,$$

This being so, it at once appears that, if (ξ, η) are coordinates of a point on the conic, then the ratio

$$(\xi - \xi^2)^2 + (\eta - \eta^2)^2 = ((\xi - 5\xi)^2 = b - 1\delta\delta)$$

which is the required condition $\Phi = 0$.

12. I consider development of the solution, writing down the equations to the conic, that there are

$$(\xi - 1)^2 - x - (Y - 5)^2 - n\delta$$

and similarly

$$\frac{x}{a-1} = -\frac{y}{a} = -1\delta - \beta - (a+b)\delta - (Y-5)^2 - n\delta$$

and hence

$$\frac{x^2}{a-b} + \frac{x^2}{b-5} + \frac{x^2}{c-5} = 0, \text{ that is, } \Phi = 0.$$

This proceeds development of the solution, the equation of the circle C being

$$x^2 + y^2 - 2\alpha x + (a^2 + b^2) = 0,$$

which is the required condition.

13. For further development of the solution, the equations of the given circles A, B, C are

$$(X - a)^2 + (Y - a')^2 = a^2,$$

$$(X - \beta)^2 + (Y - \beta')^2 = \beta^2,$$

$$(X - \gamma)^2 + (Y - \gamma')^2 = \gamma^2,$$

and if we write r, s, t for the distances of this centre from the points A, B, C respectively, then in the equations

$$r = a + 3, \quad s = b + 3, \quad t = c + 3,$$

or eliminating Σ , the centre is determined by means of the hyperbolas

$$s - r = b - c, \quad t - r = a - c, \quad t - s = a - b;$$

these three hyperbolas have, in fact, two common intersections which are the two points Z_1, Z_2 . See post No. 12 and 14.

12. In explanation of No. 4, observe that, starting from the equation

$$X_1 + iY_1 = m(X + iY),$$

and writing $mX = P, miY = iQ$, we have

$$X_1 + iY_1 = \frac{P\sqrt{1 + Q^2 \cdot 1 + k^2 Q^2} + iQ\sqrt{1 + P^2 \cdot 1 - k^2 P^2}}{1 + k^2 P^2 Q^2},$$

that is,

$$X_1 = \frac{P\sqrt{1 + Q^2 \cdot 1 + k^2 Q^2}}{1 + k^2 P^2 Q^2}, \quad Y_1 = \frac{Q\sqrt{1 - P^2 \cdot 1 - k^2 P^2}}{1 + k^2 P^2 Q^2}.$$

and thence

$$X_1 + iY_1 = \frac{P + iQ}{1 + k^2 P^2 Q^2} \cdot (\text{if } X_1 + iY_1 \text{ be put } = mP);$$

hence

$$P^2(1 - k^2 P^2)m^2 = -Q^2.$$

Now considering in figure 1 that in the rectangle parallel to the axis of X , that is, taking Y constant and therefore also Q constant, and proceeding to eliminate P , we have

$$P^2 = \frac{(1 + k^2 Q^2)m^2 + Q^2 + (1 + Q^2)m^2}{1 - k^2 m^2 Q^2}, \quad 1 - k^2 P^2 = \frac{1 + k^2 Q^2(1 - m^2)}{1 - k^2 m^2 Q^2}, \quad 1 + k^2 P^2 Q^2 = \frac{1 + k^2 Q^2}{1 - k^2 m^2 Q^2},$$

and consequently

$$X_1 = \sqrt{1 + Q^2 \cdot 1 + k^2 Q^2} \cdot \sqrt{1 + Q^2(1 - k^2 m^2 Q^2)} \cdot \frac{1}{1 + k^2 Q^2},$$

$$Y_1 = \frac{Q\sqrt{1 + Q^2 \cdot 1 - k^2 m^2} \sqrt{1 + Q^2 - 2kY(1 + Q^2)}}{1 + k^2 Q^2},$$

giving $Y_1^2 + 1 = \frac{1}{m^2}$ of these in terms of Q and m , $= X_1^2 + Y_1$. The former of these equations, replacing thereby η^2 by its value, gives

$$(X_1^2 + Y_1^2 - 2kX_1 - 2kY_1) + \frac{Q^2}{m^2} = 0,$$

viz. we have thus the equation of the curve, a bicircular quartic which in figure 2 corresponds to the line parallel to the axis of X , in figure 1.

In particular, for the line LM of figure 1 we have $Y = \frac{1}{k}$ and thence $Q = \frac{1}{km}$; that is, $Q = \frac{1}{k}$, and hence we have $\Delta = \frac{1}{k}$: the equation of the bicircular quartic is

$$(X_1^2 + Y_1^2)^2 = \frac{2}{k}(X_1 + Y_1) + \frac{1}{m^2} = 0;$$

viz. this is the circle $X_1^2 + Y_1^2 = \frac{1}{k}$ twice repeated, and we have then this circle, or rather the half-circumference LFM of figure 1, corresponding to the line LM of figure 1. More simply,

$$X_1 + iY_1 = m(\mathbf{x} + iY).$$

In all that follows, I write, as before, $\lambda - \cdot$, $r - a$, $s - b$, $t - c$, so that the last mentioned equations are then

$$r = a + \lambda, \quad s = b + \lambda, \quad t = c + \lambda,$$

and we deduce

$$\begin{aligned} r &= \frac{a^2 + \lambda^2 - a^2}{2\lambda} - \frac{1}{2}(a + \lambda - a)\frac{1}{2\lambda}, \\ s &= \frac{b^2 + \lambda^2 - b^2}{2\lambda} - \frac{1}{2}(b + \lambda - b)\frac{1}{2\lambda}, \\ t &= \frac{c^2 + \lambda^2 - c^2}{2\lambda} - \frac{1}{2}(c + \lambda - c)\frac{1}{2\lambda}, \end{aligned}$$

viz. writing

$$R = \frac{a^2 - a^2}{2\lambda}, \quad S = \frac{b^2 - b^2}{2\lambda}, \quad T = \frac{c^2 - c^2}{2\lambda},$$

and therefore

$$(R + ra) + s^2 - n^2,$$

these equations are

$$\begin{aligned} r &= R + \frac{1}{2}n, \quad s = T - \frac{1}{2}n, \quad t = R + \frac{1}{2}n, \\ &= T + n, \quad s = R + \frac{1}{2}n, \end{aligned}$$

R, S, T are each of them a linear function of the coordinates (x, y) , say we have

$$R = ax + a'y + b'a,$$

$$S = mx + n'y + p,$$

$$T = mx + n'y + p,$$

where

$$\begin{aligned} \lambda, \lambda_n &= \frac{B - \gamma}{r}, \quad \frac{B - \gamma}{r}, \quad \frac{(B + \beta) - x' - 4' a'^2 - 4'^2}{-2} \\ \mu, \mu_n &= \frac{\gamma - a}{s}, \quad \frac{\gamma - a}{s}, \quad \frac{\gamma^2 - x^2 - a'^2 - 4'^2}{4} \\ \nu, \nu_n &= m', n', \quad \frac{a - \beta}{t}, \quad \frac{a^2 - x^2 - b'^2 - 4'^2}{2} \end{aligned}$$

13. From the two equations $r = R$, $s = T - \frac{1}{2}\lambda$, we deduce the equations of a line and a circle.

The line is $R - T - \frac{1}{2}(b + b) = 0$, viz. substituting for R and T their values, this is

$$\frac{\beta\gamma - 4}{2\beta} = \frac{2\beta - \mu)(a - \alpha)}{\alpha} = (m - n)x = 0,$$

that is,

$$a(\beta - 1) - n(\beta - 0)y + 2\Gamma = 0,$$

or, since $1 + m = a + 0$, the equation is

$$3a^2 + na^2 + ay + 4na = 0.$$

which is symmetrical in regard to the three circles. The equation may be written

$$\frac{l(s^2 - a^2) + m(r^2 - b^2) + n(s^2 - r^2)}{2(a + ms + n)} - (am + 3n)\beta - (a + b)\beta + a^2 = 0,$$

and it then appears that the line passes through the radical centre of the three circles.

We have

$$(s - \rho^2)r = a^2 - \delta + 5b - \beta s + s^2 - r^2 = a^2 + \beta r + s + \beta n + b\beta - \beta + \beta r - r),$$

$$(t - \rho^2)b - \rho^2 r = a^2 - \delta + 5b - \beta s + s^2 - r^2 = a^2 + \beta r + s + \beta n + b\beta - \beta + \beta r - r),$$

and thence

$$(c - 5)t^2 - a^2 = (s^2 + \beta^2 + 1 - 5)b\beta\beta\beta = 5)r + (s + \beta r) - (r + \beta - r)$$

after some easy reductions the equation is found to be

$$8(K' + K')[K^2(sr - \beta) - 2\delta b - \beta] - 2(\delta\beta + \beta) - \delta + \beta - \delta + Y, \beta\delta b\beta] = 0,$$

$$\delta = 5(R + K, K_n)[K' - 3]\beta - 5(\xi K' + K' - K, K_n) + (\xi K' - K_n - 5)\delta b\beta,$$

14. To further abbreviate, I write

$$(B - y)x + (a - y)y + b\beta\beta\delta = F,$$

$$(B - y)x + (a - y)y + b\beta\beta\delta = -F,$$

$$(B - y)x + (a - y)y + b\beta\beta\delta = -F,$$

also,

$$[\delta r + ay + 4(B(F))] = \xi(B + ay + b\beta + b\beta + \beta) - H_n,$$

and then writing down these three equations, we have

$$8(K' + K')[\nu^2 - 2[\Pi + b\beta + a^2 + 2 + (b\beta + \beta + b\beta) + ay + \beta\beta]],$$

$$8(K_n + K')[\rho - 4\Pi + bF + \delta + (b\beta + \beta + b\beta)F + ay + \beta\beta]],$$

$$8(K + K')[\rho - 4\Pi + bF + \delta + (b\beta + \beta + b\beta)F + ay + \beta\beta]],$$

which are the equations of the three Newton-circles, each touching the chord

$$A + ay + b\beta y + ay + 4\beta\beta - 0.$$

15. The first of these equations is

$$4(K' + K') \cdot a' - 2(F + b\beta + b\beta) + b\beta)$$

$$= 2(a(K' + K')\nu - 5\nu + b \cdot 2II(B' + F'\beta\beta),$$

$$= b(K' + K')\nu + (\beta)\beta - 2(K + ay + b\beta + b\beta + b\beta + b\beta + F),$$

$$= 4(K' + K')\nu - 8(b\beta - 2(\beta - 2(\Pi + b\beta + b\beta + F + b\beta + F),$$

that is

$$\begin{aligned}
& [4(K' + K') - 2(\beta - \nu) + h(B + \nu) - 2(\Pi + b\beta + b\beta + b\beta + b\beta + F)] \\
& = 4(K' + K')\nu + 2[(\xi(B + h(B + b\beta + b\beta + b\beta + b\beta + b\beta + F))), \\
& \quad + 1 - 5(K' + K')\Pi^2(B + b\beta + F), \\
& \quad + 1 + 5(K' + K')\Pi^2(B + b\beta + F), \\
& \quad + 1 + 5(K' + K')\Pi^2(B + b\beta + F),
\end{aligned}$$

where the last term is

$$\begin{aligned}
& = 16(K' + K')\Pi^2(B + b\beta + F), \\
& \quad + 4(K' + K')\Pi^2(B + b\beta + F), \\
& \quad + 16(K' + K')\Pi^2(B + b\beta + F), \\
& = (R' + R')[\nu - 4(B + F)\beta + 4(F)P - 4 + 16\beta + (\delta + F), \\
& \quad = -16\beta + 1(\beta\beta + 4\beta) + 4F)(\beta\beta\beta + (\beta\beta + 4\beta) + 4F))
\end{aligned}$$

It thus appears that the equation of the Newton-circle F is

$$\begin{aligned}
& 4(K' + K') - 2(F + b\beta + b\beta) + b\beta + b\beta + b\beta + b\beta + b\beta - 2(\beta + \nu^2 + \alpha + \beta^2) \\
& = (K' + K')[\nu^2 + 4(F + b\beta + F) - 2(F + \beta b\beta + \beta) + b\beta + 4 + b\beta + b\beta + 4 + b\beta - 2(\beta + \nu^2 + \alpha + \beta^2)]
\end{aligned}$$

where the coordinates of the centre are

$$\begin{aligned}
i_n &= \frac{2(K' + K')\nu - (1 - 5(K_n + F))}{2(K' + K - 1)\nu^2}, \\
i_n &= \frac{2(K' + K')\nu - (1 - 5(K_n + F))}{2(K' + K - 1)\nu^2},
\end{aligned}$$

and

$$\text{rad}^2 = \frac{(\xi + K)\nu^2 + 4(K_n + K_n)}{4(K' + K' - 1)\nu^2}((\Pi + F - 2(B)^2 + (\xi + F + 4\beta)\nu),$$

and similarly for the Newton-circles G and H .

16. The centres are in a line at right angles to

$$= 2K_n + 2K_n + a + b\alpha = 0,$$

say the equation of this line is

$$K'x - K'y + \nu = 0.$$

This must equally be verified

$$\frac{K_n^2(K_n + K')\nu - (1 - 5(K_n + F))(\xi + K' + K' - K_n + (1 - F)F)}{2(K' + K - 1)\nu^2} + \nu = 0,$$

that is,

$$2(K' + K')(K_n - K_n) - (K_n + K' + K'(K_n + K')\nu - K'(F - F) - K_n^2 + (K_n - F)\nu = 0,$$

viz. we ought to have

$$2(K' + K')(K_n - K_n) - (K_n + K' + K'(K_n + K')\nu - (K_n - F) + K_n^2 + (K_n - F)\nu = 0.$$

This should agree with

$$2(K_n + K')(K_n - K_n) - (K + K' + K'(K_n + K')\nu - K(F - F + (K_n + K')\nu = 0.$$

viz. we ought to have

$$2(K_n + K')(K_n - K_n) - (K_n + K' + K'(K_n + K'))\nu - K(F - F + (K_n + K')\nu) = 0.$$

This can be true only if $K_n(K_n - K) = (K(B - \beta) - K_n(F - F))\nu$, and, in fact, the

$$\begin{aligned} K_n(K - K_n) &= (R(B - \beta)) - K_n(F - F)) - K_n(F - F)), \\ &= 2\sqrt{K + K_n}, \quad \nu = (K + R - F) + K(F - F), \\ &= 2\sqrt{K + F} - (\nu - \alpha + b - b\beta\beta, -F) - F - F + (\nu - F) \\ &= -\nu(\beta - 0\nu - \nu + n\beta + \nu - 0\alpha)((B - F)), \end{aligned}$$

and hence

$$\begin{aligned} F - G + \nu(F - K, -F), \\ -K, F - G, -2K + F. \end{aligned}$$

and thus the equation is verified.

17. Writing for shortness

$$3y_n - 2(y, \nu n - n\nu, \nu(n - \nu - 2(\nu, X, Z), N - n - \alpha - \alpha - \alpha,$$

we have

$$\begin{aligned} 2(K' + K')(R' + Z) + (Z - F)(R + Y) - K(R + X + Y + Z) \cdot W &= 0, \\ +\alpha((F - \alpha)\nu + (Z - \alpha)Y) + \alpha\alpha + \beta(Y - Z + X + Y - Z) \cdot W &= 0, \\ +\alpha((R - \beta)Z + (X - \beta)Y) + \alpha\alpha + \beta(Y + (Z - \beta)\Pi) \cdot W &= 0, \end{aligned}$$

To verify this, observe that the left-hand side is

$$2(R + K_n)(\alpha + Z) + (R - Y)\alpha + \alpha(X - Y - Z) - \alpha(X - Y - Z),$$

or putting hence $X + Y + Z$ for this

$$\begin{aligned} &= \alpha\alpha((B - \alpha)X + (Y + (Z - \alpha) + \alpha(b - \alpha)KZY))Y + Y + Z, \\ &\quad +\alpha((\gamma - a)Z + (\alpha - \alpha)Y), \end{aligned}$$

This equation determines Φ , and thus the equation of the line of centres is

$$2(K' - K')(KX - K_n)Y + (KH + K, H_n) - \Phi = 0.$$

18. This line meets

$$-2K_n + 2K_n + 0 = 0,$$

in the mid-point of the chord Z_1Z_2 . We thus have for the coordinates x, y of this mid-point

$$2(K' + K')(K' - K')X + 2(K_n)Y - (KH + K) - K(K' + K' + K_n)(\alpha + b + b\alpha + b\alpha + b\alpha) = 0,$$

$$2(K' + K')(K_n)Y + 5(Y) - 2((KH + K, H_n) - 1)(K' + K' + K_n)(b + b\alpha + b\alpha + b\alpha) = 0.$$

The perpendicular distance of the centre F from the chord Z_1Z_2 is

$$= \frac{(KH + K' - 1)(\xi + 4 + (F - \xi(\beta + a) + (1 + F + -2a\alpha))Y)}{2\sqrt{K + K'}}.$$

We have

$$K' + K' - \nu = (\alpha\alpha + \alpha\delta + ay)(s + r) + a + a)(\xi + \beta)\beta + 5y - ny + a) - 4(R)\nu_n,$$

but

$$J^2 = a(\xi - y)\nu + (a' - y),$$

$$g^2 = \alpha^2 + (\beta - y),$$

and hence

$$\begin{aligned} -f^2 + g^2 + h^2 &= a^2 - (\xi - 4 + \alpha) - 4 + (\xi - 4\alpha)^2 + ((\xi - 4)^2 - (\xi - 4)^2), \\ &= -f^2((\xi + y)\beta + \alpha)(\xi + (\beta - y))^2 = a^2 + \alpha^2 - (\xi - 4\beta + 4 - 4) \end{aligned}$$

or forming the like values of $-g^2(f^2 + g^2 + h^2)$, and $-h^2(f^2 + g^2 - h^2)$, and adding, we have

$$\begin{aligned} \Delta &= f' + g' + h' + \alpha' + 5)\nu + 2(2\alpha + \alpha\beta\beta - 4y)b - 4(\alpha)\beta\beta b, \\ &\quad + \alpha(\delta b - \nu)\xi + 4)\beta + 4\beta\beta)b, \\ &\quad + Y(\beta\beta a - \alpha\beta - 4\beta - 4\beta\beta\beta b - 4(\alpha)\beta\beta b), \\ &\quad - \alpha(\delta y + (\xi + b)\beta + +(\alpha + \beta) + +Y(R)\delta - 4D. \end{aligned}$$

But

$$\nu\delta^2 + \mu y + \nu y + 4\beta\delta - 4\beta \cdot \xi - 4\beta \cdot \xi - 5\beta - 5(\beta\beta - \beta) - 4(\Pi - \mathcal{L}),$$

that is,

$$-4\nu + \alpha\delta + ay + 4b = -2 + 2(\Pi - \mathcal{L}),$$

or say

$$\delta\nu + \alpha\delta + ay + 4b = -\nu(\Pi - \mathcal{L}).$$

Thus

$$4(K_n + K_n) - \nu' + (4F)b + 4\beta b = -4D''\nu - \Delta,$$

and therefore

$$4(K' + K') - \nu^2 = 4(a\nu)^2 + 4\beta b\delta;$$

But

$$\begin{aligned} \Delta(K + K) &= -\nu, -4\beta b - F), \\ \delta(K + K') - F^2 - \nu^2 &= -4(\beta - F); \end{aligned}$$

and hence

$$\begin{aligned} \delta(K + K) - F^2 &= -4(\beta - F), \\ \delta(K_n + K_n) - F^2 &= -4(\beta - F), \end{aligned}$$

hence

$$\frac{K + K_n}{(K' + K' - 1)\nu^2} = \frac{-40}{8\Pi^2}.$$

and we have

$$(KZ)_n = \frac{-46}{2\nu^2}[(\Pi + F - 2\beta(F))^2 + (\xi + F - F)b\beta - 2(\Pi - F)b\beta + 4n^2].$$

20. This should agree with the expression in No. 7, that is, we ought to have

$$(\Pi + F - 2\beta(F))^2 + (\xi + F - 4\beta b\beta) - 2(\Pi - F)b\beta + 4n^2.$$

and this breaks up into the equations

$$\begin{aligned}(\Pi + 2\beta b\beta) + 2(\Pi - F + 4\beta) &= f^2 g^2 h^2, \\ 8F^2 - 2\beta b\beta(F + 2) - (\xi + F + b\beta + 4\beta(\Pi - F + 4\beta) + 4b\beta b\beta b\beta) &= -4nF + (F - F), \\ F^2 - 4y(F)^2 &= 2(F + 8\xi b\beta) + b\beta - 2(\beta + 4\beta(\Pi + F + b\beta + 4)), \\ -P\beta(R)^2 &= -F\beta\beta,\end{aligned}$$

which may be separately verified.

21. In No. 4,

$$\begin{aligned}(B - y)X + (s - y)y + (s - z)y - z &= 6 + (8 - 1)\nu - 6\delta(F), \\ &= (B - y)X + (8 - y)y + (s - z)y - z = 6 + (B - F)\nu - 6\delta(F). \\ &= -6(\delta - y)X + (\xi - y)y + (s - z)y - z = 6 + (B - F)\nu - 6\delta(F). \\ &= \alpha(\beta - y)\nu + (\xi - y)y + (s - z)y - z + (Y - z)b\beta, \\ &= -\alpha(\beta - y)\nu + (\xi - y)y + (s - z)y - z + (Y - z)b\beta,\end{aligned}$$

or, putting for shortness $\nu = \alpha + 8(s + a + ay) + \alpha + 4b\beta\alpha + b\beta + b\beta + b\beta + a$, $B = y - x$, $a = 4 - z$, &c., have a meaning different from that assigned to them in No. 7, this is

$$H + 2uy\beta = \lambda\mu p + 4p^2,$$

and similarly

$$1 + 5u \cdot y\delta + svy\beta\nu py\beta y\beta - +y\delta\beta y\nu - rvy\beta\nu + r\nu py,$$

Also,

$$\rho^2\beta = (\xi - z)x' + 5\beta - z)y'\beta + r\beta =;$$

and the first equation thus becomes

$$(-4y\alpha - 4\beta)\nu + (4(s - 4)b\beta + \alpha)\beta + (\xi - 5\beta) - 0 - \nu)X + (Y - y)x - \beta)Y - (Y - \beta)Y,$$

(when the letters $X, y, Z, w, \mu, \nu_n, \nu$, have a meaning different from that assigned to them in No. 7), this is

$$\lambda = (s - z)y\beta\nu ry(2 - y)y\beta = -2\rho y u' \nu y - y\nu py,$$

which is right.

22. For the second equation, we use the values

$$\begin{aligned}\Pi - 2u\xi\nu - (\alpha\beta + by)\nu + (\alpha\beta + \beta y)\nu, \\ H - 2u\beta - 2((\alpha + a)y + (a' - y)\nu)\end{aligned}$$

and the equation to be verified thus is

$$\begin{aligned}2F + (ar + ay)\nu - ((\alpha\beta + by)\nu + (\alpha\beta + \beta y)\nu) \\ + 5F + a(ay + ay)\nu(\nu - 5(B - y)\nu^2 + (\xi - y)\nu)X \\ - 5b - 2(K' - K\nu), \\ - 5K \cdot 2(K\beta + K\delta)\nu,\end{aligned}$$

we have

$$F = mb + (m - r)\nu + b(\xi)\nu + (\xi - y)\nu)y + (\alpha + 4)y + b)b\nu,$$

that is,

$$F = mb + (a' - y)y + b(\xi)\nu + (\xi - y)\nu + (\xi + y)y + (\xi + 4y)b\beta + b\beta,$$

and similarly

$$F = mb + (a' - y)y + b(\xi)\nu + (\xi - y)\nu + (\xi + y)y + (\xi + 4y)b\beta + b\beta.$$

Also,

$$\begin{aligned}\Theta &= 2(K + K\beta). \\ &= -1(aY(X + aY)(\xi - y)\nu - 2 \cdot 2(\xi a + 5y) + \alpha\nu(\beta y + \alpha)\nu + (\xi a + 5y))\nu, \\ &= -F(aY(X + aY) \cdot (\xi - y)\nu y - 5 - 2 - \beta)(\beta a + \beta y)(5y + \beta y))y - 2b\nu + 5y), \\ &= -0(aY(X - 0(aY - 5b)\nu) - 2(\alpha + ay)(aa + 5y + (\beta + y)5 + 5y)).\end{aligned}$$

The left-hand side is

$$\begin{aligned}&= -2((a + b)y + (a + b)y)(aa + 5y) + 2((b + a)\nu)\nu + (a + b)y), \\ &\quad -2(0 - 2(\alpha y(F + 2(\alpha + b))y), \\ &\quad -0(aY + ay)(\xi + aY) + (a + ay)(\nu + ay) + 2(\xi y + b)y), \\ &\quad -2(b + ay)(aa + 5y) + 2(b + ay)\nu + (a + b)y), \\ &\quad -a(Y + ay)(a + 4)\nu + (a + b)y),\end{aligned}$$

which is equal to the right-hand side.

23. For the third equation we have as above

$$F^2 - \alpha y\beta - 2y\nu y, F_n = -4y - 4y - \alpha\nu y,$$

$$\Theta = 2(K_n + K + K_n) = -4y(F + 4y) + ay),$$

and the equation thus is

$$(2u\nu + \nu y)(\nu + ay)\nu + 5y) - \text{Bina} + 5(\xi(\nu y + \nu y) + (aY + ay)(F));$$

here the left-hand side is

$$\begin{aligned}&= (\xi - 5b\nu(\xi + 4)) \\ &\quad + (\alpha - 5b\nu(b + aY)) \\ &\quad + 2\nu y\beta(F + ay) + \alpha)\nu - 4y + (ay + 4y)) \\ &\quad + 5\nu(F + 4y) - F + (aY + ay)F,\end{aligned}$$

which is equal to the right-hand side.

24. The fourth equation is the identity $-F\nu ny - \nu F\nu y$, and the whole equation is thus verified: viz. the analytical solution leads to the expression

$$\Phi(Z_1 Z_2)^2 = -4(f^2 - (b - c)^2)(g^2 - (c - a)^2)(h^2 - (a - b)^2)\Phi,$$

obtained by a geometrical process in No. 7 for the squared distance of the two centres.

ON ORTHOMORPHOSIS

[920]

From the Quarterly Journal of Pure and Applied Mathematics, vol. XXV. (1891), pp. 203—226.

1. The equation $x_1 + iy_1 = \phi(x + iy)$, where ϕ is in general an imaginary function (that is, a function with imaginary coefficients), and where $ix, y_1, (x, y)$ are rectangular coordinates, determines, x_1, y_1 each of them as a function of (x, y) , and hence, when eliminating x, y , we have a set of curves depending on the parameter x_1 and secondly eliminating x , we have a set of curves depending on the parameter y_1 ; we have thus two sets of curves, or curve families, depending on the parameter the two sets of lines $x = \text{const.}$ and $y = \text{const.}$ respectively. They thus correspond at right angles, and are moreover such that, going to the parameters values of equal infinitesimal intervals (the same for x and y respectively), they form a doubly system of orthomorphic curves; say their case is small enough such that they form everywhere two trapezia of one of the curve families and one of the other) is a fundamental rectangle. We may assume in advance a tracing of X , and also the corresponding system of XY ; we have then in order to determine the relation between the successive trapezium X and Y , we may with which to compare it; and the successive trapezium $X, Y, Y, \&c.$, respectively, and forming with the first ones of the same the value of $f(x)$ along the double system of infinitesimal squares.

2. But to translate this into Analysis, and to obtain the equations of the two sets of curves, we proceed as follows:¹

Suppose that the curve S is given in the form

$$x_1 = p_1, \quad y_1 = q_1,$$

where p_1, q_1 are given real functions of a variable parameter θ ; we obtain a series of separate points of S and T , one of the first set being the given curve S , by forming from these equations the equation

$$x_1 + iy_1 = p_1 + iq_1,$$

and writing therein $\theta = \alpha + i\beta$. In fact, for the value of the complete parameter $\alpha + i\beta$, we have from these equations the equation

$$x + iy = p + iq,$$

where p, q are the real and imaginary parts of a known function of $\alpha + i\beta$, so that the elimination of α from these equations leads to an orthomorphic curve S' . We assume that this curve S' is given in the form

$$x = p_2\theta = \theta_2,$$

and by this is determined a third curve Θ , in the same connective curve T , on which Θ may be assumed at pleasure. Say we have

$$Cw = jCw \frac{d\theta}{p^2 + q^2} + B,$$

where p^2, q^2 are the derived functions of p, q , in regard to θ ; C is a real constant, the value of which may be assumed at pleasure. But we have

$$ww + xy = w' + x'y' = w' + iy,$$

which is the same thing as $w = u + iv, \&c.$

¹The investigation is taken from the memoir, Schwarz, "Ueber einige Abbildungsaufgaben," *Crelle*, t. LXX. (1869), pp. 105—120.

3. We may, in the equations for w , u , substitute for θ any other real value θ_n , whatever; say the equations then become

$$\theta = u, \quad \theta - u + iy, \quad \theta = u + iv, \quad \text{values which give}$$

$$x_1 + ip_1 \cdot y_n + 2y_n + (p_1 + iy)d_n.$$

4. We very, in the expression for w_1 , w_2 , substitute for θ any other real value θ_n , whatever; say the equations then become

$$w + a = u, \quad w - u + iy, \quad w = u + iv,$$

which give

$$w_1 = (p_1 + iy_1) + iy(p^2 - 4i^2).$$

the proof consists in showing that there exists for θ , θ_n values, differing infinitesimally from θ and such as to reduce this equation to

$$x_1 + iy_1 = p + i\dot{q}_1 + (p^2 + iy^2)\frac{d\theta}{d_n},$$

the value assumed by $p + i\dot{q}_1$ on substituting for θ this value is $\theta_n + \frac{d\theta}{d_n}$; for this being so, it is obvious that we have for S' the curve in the series of curves S which corresponds to the value $\frac{4}{n}$ of θ in $w = w + a$. The value of θ , in this form

$$\theta_1 = \theta + a, \quad p^2 + a)y + (q^2 + ay)\frac{d\theta}{d_n},$$

$$\theta_2 = \theta + a, \quad p^2 + a)y + (q^2 + ay)\frac{d\theta}{d_n},$$

and since $p\theta + a$ is infinitesimal, we may in the terms $p\theta'$, and $\dot{q}\theta'$, write θ_n in place of θ , and so reduce this to

$$x_n = p\theta + a)p^2 + ay\frac{d\theta}{d_n}, \quad y_n = q\theta + a)q^2 + ay\frac{d\theta}{d_n},$$

or, for $p^2\theta + ay\theta$ substituting in this $\frac{1}{n}\frac{d\theta}{d_n}$, this gives

$$x_1 = p + y\beta + \alpha + (p^2 + iy\beta)\frac{1}{n}\frac{d\theta}{d_n},$$

which is the before-mentioned equation.

The following two examples are given by Beltrami.

4. First, let the curves S , S' be given confocal ellipses,

$$\frac{x^2}{a^2 + p^2} + \frac{y^2}{p^2} = 1, \quad \frac{x^2}{a^2 + q^2} - \frac{y^2}{q^2} = 1, \quad \text{thus } a^2 = e^2.$$

Here

$$p = \alpha \cos \theta, \quad q = b \sin \theta,$$

whence

$$p^2 + 4q^2 = a^2 \sin^2 \theta + b^2 \cos^2 \theta,$$

$$F = \frac{C}{2} \cos \theta, \quad Q = b\beta\theta,$$

whence

$$y^2 Q = y^2 \int \frac{d\theta}{a^2 \sin^2 \theta + b^2 \cos^2 \theta},$$

and consequently

$$Cw = \int \frac{d\theta}{a^2 \sin^2 \theta + b^2 \cos^2 \theta}, \text{ or taking } C = \frac{c}{ab}, \text{ then } w = R,$$

and for the two sets of curves we have

$$x_1 + iy_1 = a \cos(u + iv) = a(\cos u \cosh v - i \sin u \sinh v),$$

which are two sets of confocal curves. In verification, write

$$\cos u = e^X, \quad \sin u = e^X, \quad \text{whence } X^2 = e^{X^2} - 1,$$

and that is,

$$x_1 + iy_1 = a(X + iY - (X^2 - 1)(X^2 + 1)),$$

that is,

$$x_1 + iy_1 = a(X + iY - CX^2 + (X^2 + 1));$$

and consequently

$$\frac{a^2}{X^2 - 1} + \frac{a^2}{X^2 + 1} = -2\alpha + 0\beta, \text{ which the curves } T,$$

where observe that,

$$(aX + bY)^2 - (bX + aY)^2 = -4(aX - bY),$$

so that, putting $(aX + bY)^2 = e^{ik}$, we have

$$(bX + aY)^2 = e^{ik} + 4 = (e^{-k} - 4)^2 = -4e^{2k},$$

and thus the curves S are the confocal ellipses

$$\frac{x^2}{a^2 e^{2k}} + \frac{y^2}{b^2 e^{2k}} = 1, \quad -k < c < \infty.$$

and similarly $X^2 + Y^2 = 1$, putting $(bX + aY)^2 = e^{i\nu}$, we have

$$(aX^2 + bY^2)^2 = (e^{i\nu} - 4)^2 = -4e^{2\nu},$$

and thus the curves T are the confocal hyperbolas

$$\frac{x^2}{a^2 + b^2 e^{2k}} + \frac{y^2}{b^2 + a^2 e^{2k}} = 1, \quad -k < c < \infty.$$

5. Next, let the curves S be a circle and the curve S' an inscribed concentric circle. It is convenient to take the circles such that they have for their common chord the line $y_n = 0$ instead of the more usual $a_n = 0$, the points $a_n = \pm$ being the points of contact:

$$\frac{x^2}{a^2 + b^2} + \frac{y^2}{b^2} = 1, \quad -k > c > 0.$$

This being so, the equations of the two circles are

$$x_1^2 + y_1^2 = 2p, \cos \mu, \quad p_1 = 1,$$

$$x_n^2 + y_n^2 = 2p, \cos \mu(1 - y \cos \mu) = -1,$$

and we have

$$p = \cos \mu \cos \theta, \quad q = \cos \mu \sin \theta + \cos \mu,$$

$$P = -\cos \mu \sin \theta, \quad Q = \cos \mu \cos \theta = \cos^2 \mu - \cos \mu \cos \theta,$$

giving

$$p^2 + q^2 = \cos^2 \mu,$$

$$y'cdw = \frac{d\theta}{\cos \mu \sin \mu} = \cos \mu \int \frac{d\theta}{1 - 4 \cos \mu \cos \theta + \cos^2 \mu}, \text{ then } dw = \frac{1}{\cos \mu} \cdot \frac{d\theta}{1 - 4 \cos \mu \cos \theta + \cos^2 \mu},$$

The integral of this may be written

$$\tan(jw - \pi) = -\cot \frac{1}{2}\mu \tan(j\theta + \frac{1}{2}\pi),$$

and if we have assume for the constant of integration p , a value such that we find for α the equation

$$\tan \frac{1}{2}n = \cot \frac{1}{2}\mu \tan \cos \mu,$$

we have $w = +b$, and

$$x_1 + iy_1 = \cos \mu (x_n + iy_n - \cos \mu \cos \mu) = \cos \mu \frac{1 + i \tan \beta(n + iv)}{1 - i \tan \beta(n + iv)},$$

hence

$$x_1 + iy_1 = \cos \mu \cdot \frac{-(b + b^{-1}) + i(a + a^{-1})}{1 + ib - ab}, \text{ then, } dw = \frac{a(a + a^2)}{1 + a^2 + ba^2},$$

or writing, as we may then $2a, b$ in place of a, g , respectively, then $a_n = i\beta a$, where

$$a_n = \frac{-a + a(1 + b^2) - a(1 - \beta^2)}{1 + a(1 + \beta^2 + a^2 + b^2)}, \quad y_n = \frac{a(1 - a^2 - b^2)}{1 + a(1 + \beta^2 + a^2 + b^2)},$$

$a_n - a_n -$ for the curves S and T respectively. For the curves S then $\Gamma = \cot \frac{1}{2}\mu$ gives the original S -curve

$$a_1^2 + y_1^2 - 2\Gamma \cos \mu = -1.$$

We have thus the well-known system of orthotomic circles, those circles of the one set pass through the points $a_n = \pm 1, y_n = 0$, and those of the other set through the antipodes $a_n = \pm 1, y_n = 0$.

6. A different solution is given in the Dissertation referred to below²: I reproduce this, giving the proof in a somewhat altered form. The equations of the curves S are: $\phi(x, y) = 0$, $\phi(x, y) = a$, $\phi(x, y) = a$, etc.

In the investigation, we introduce the conjugate variables $z = x + iy, \bar{z} = x - iy$, and write Δ, ϕ , to denote values assumed by the functions $\phi(x, y), \psi(x, y)$ when x, y are replaced therein by their values in terms of $z, \bar{z}$, viz. $x = \frac{1}{2}(z + \bar{z}), y = \frac{1}{2i}(z - \bar{z})$. But in the first instance we may, without attending to the meanings of $z, \bar{z}$, consider ϕ, ψ , as denoting each of them a given function of the two co-ordinates $z, \bar{z}$.

Suppose now that the functions $f(z), f_1(z)$ are determined by the equations

$$f(z) = \int \frac{1}{\phi} \cdot \frac{d\phi}{dz} dz, \quad f_1(z) = \int \frac{1}{\phi} \cdot \frac{d\phi}{d\bar{z}} d\bar{z},$$

where in the first integral, after the calculation of $\frac{d\phi}{dz}$ is expressed as a function of $\bar{z}$ by means of the second integral after the calculation of $\frac{d\phi}{d\bar{z}}$ is expressed as a function of z by means of the equation $\phi = 0$; and where in each integral the constant is properly determined, as presently appearing.

²Notes, Ueber die ten gemein lassen and zur die gleichseitigen polischen Aufgaben Riemann zur Lösungs, Inaugural Dissertation der Universität zu Göttingen, vol. XXV, 1869. I quote this in full as the explains the object of the present memoir.

We have

$$f'(z)dz + f'_1(\bar{z})d\bar{z} = \frac{1}{\phi} \left(\frac{d\phi}{dz}dz + \frac{d\phi}{d\bar{z}}d\bar{z} \right) = \frac{1}{\phi}d\phi,$$

where the expression on the right-hand side is $= 0$, if only $\phi = 0$; that is, ϕ being $= 0$, we have $f'(z)dz + f'_1(\bar{z})d\bar{z} = 0$, and consequently $f(z) + f_1(\bar{z}) = \text{const.}$, viz. this last equation, existing as a consequence of $\phi = 0$, is equivalent to the relation $\phi = 0$.

Writing now $z + iz$, $\bar{z} - iz$ in place of z , $\bar{z}$ respectively, where iz , $\bar{z}$, are arbitrary infinitesimal increments, we have

$$f(z + iz) + f_1(\bar{z} - iz) = 0,$$

equivalent to

$$\phi(x + iz, y + i\bar{z}) = 0.$$

that is,

$$f(z) + zf'(z) + f_1(\bar{z}) - izf'_1(\bar{z}) = 0,$$

equivalent to

$$\phi + iz\frac{d\phi}{dz} + iz\frac{d\phi}{d\bar{z}} = 0;$$

and if we have $\phi = 0$, then

$$iz = zf'(z),$$

$$iz = zf'_1(\bar{z}),$$

then we have

$$f(z) + f_1(\bar{z}) - 2z = 0,$$

equivalent to

$$\phi = \frac{1}{2f'(z)} \cdot \frac{d\phi}{dz} + \frac{1}{2f'_1(\bar{z})} \cdot \frac{d\phi}{d\bar{z}} = 0,$$

that is, in virtue of

$$f'(z) = \frac{1}{\phi} \frac{d\phi}{dz}, \quad f'_1(\bar{z}) = \frac{1}{\phi} \frac{d\phi}{d\bar{z}},$$

equivalent to

$$\phi - 2z\phi = 0.$$

There is the difficulty that the last-mentioned values of $f'(z)$, $f'_1(\bar{z})$ do not subsist absolutely, but only when $\phi = 0$; the answer to this is that the equation $\phi - 2z\phi = 0$, where z is now a real variable, instead of the one used previously, z is identified by the equation $\phi = 0$, but considering for a moment that $\phi - 2z\phi = 0$ as connected by the equation $\phi - 2z\phi = 0$, then the value of ϕ .

Instead of being actually equal to $2z$, will be equal to $2z$ plus an infinitesimal value containing the factor ϕ , and consequently on substituting this value in the equation $\phi - 2z\phi = 0$, and neglecting powers of z , we obtain the expression $\phi - 2z\phi$.

If we now write $z = a + i\beta$, $\bar{z} = a - i\beta$, we obtain the expression $\phi(x, y) = z$ a real quantity; and the equation $f(z) + f_1(\bar{z}) - 2\alpha = 0$ is therefore the equation of an orthomorphic curve S .

Considering this equation $f(z) = a + i\beta$ as that denoting the two equations, and successively eliminating the parameters z and $\bar{z}$, we obtain a system of curves S depending on the parameter a , and secondly a system of curves T depending on the parameter z , which are two systems of orthomorphic curves. And by what precedes, it appears that the curve S belonging to the value $a = 0$ is the same as the consecutive curve S' belonging to a real infinitesimal value $z = \nu$, are the given curves

$$\phi(x, y) = 0 \text{ and } \phi(x, y) = -2x\phi(x, y),$$

respectively.

7. For instance, let the curves S and S' be the circles

$$x^2 + y^2 - 1 = 0, \quad x^2 + y^2 - 1 - 2ax = 0,$$

here

$$\phi = xz - 1, \quad \phi = j(z + \bar{z}),$$

$$\tan \frac{1}{2}u - \tan^{-1} \alpha + \tan^{-1}(z + i\alpha). \text{ It we may write,}$$

$$\frac{1}{z} \cdot \frac{dz}{d\bar{z}} = \frac{u + i\beta}{1 - u i\beta}, \quad \frac{u}{z} \cdot \frac{1 + i(z + i\beta)}{1 - i\beta(z + i\beta)} = i \log \beta = u,$$

that is,

$$z + iy = \cot \beta \cdot \tan \tan u,$$

giving

$$x + iy = \frac{u(1 - i\alpha)}{1 - i\beta} = \frac{\beta(1 + a^2)}{1 + a^2\beta^2}.$$

The two sets of curves are then given by

$$z + iy = \tan^{-1}(z + i\beta), \quad \tan^{-1}(z + i\beta) - \tan^{-1} \alpha + (z - \beta(z + i\beta) - 2\delta,$$

viz. as the constant of integration, this is

$$x + iy = \tan^{-1} - 4\beta\nu, \quad \tan^{-1}(z + i\beta) - \tan^{-1} \alpha + (z - \beta(z + i\beta) - 2\delta.$$

The two sets are of curves are

$$x + iy = \cot \beta\nu(1 - i\beta),$$

that is,

$$x + iy = \frac{u + i\beta + 4(1 - i\beta(z + i\beta) + 2)}{1 + u(1 + i\beta)},$$

that is,

$$x + iy = \frac{u + i\beta(z + i\beta)}{1 + u(1 + i\beta)} = \frac{u + i\beta(z + i\beta)}{1 + u(1 + i\beta)},$$

giving

$$x^2 + y^2 - 1 - 2 \left(\frac{1 - a}{a} \right) = 0,$$

for the two systems of curves. The first of these, substituting for a its value, becomes

$$x^2 + y^2 - 1 = 2ax,$$

and thus the curve, for $a = 0$ and a determinate value, are

$$x^2 + y^2 - 1 = 0 \quad \text{and} \quad x^2 + y^2 - 1 - 2xa = 0,$$

as they should be.

8. Reverting to the first-mentioned scheme, the equation of the circle $a^2 + y^2 - 1$ is satisfied by $x = \cos \theta$, $y = \sin \theta$, and we have therefore $x_1 = i\alpha^2 e^{i\nu}$; and hence by what precedes, writing for convenience $X + iY$ instead of $a + i\alpha$, we have

$$x_1 + i\beta_1 = ae^{i(X+iY)},$$

and the two equations give $x_1^2 + y_1^2 - 1 = 0$; the two circumferences thus correspond to each other. It is to be noticed that, if the numerator function contains a factor $(x + iy)^n$, say if $\phi(x + iy) = (x + iy)^n \phi_1(x + iy)$, then the denominator contain a factor of the form S . Now,

14. The proof depends on the following lemma:³

For the given area F , it is possible to find, and that in one way only, a real function ξ of the coordinates (x, y) satisfying the partial differential equation

$$\frac{d^2\xi}{dx^2} + \frac{d^2\xi}{dy^2} = 0,$$

throughout the area, and which, for the values (x, y) belonging to the contour S , has a value which is a given function of x, y .

(1) ξ is throughout the area finite and continuous; except only that in the neighbourhood of a given point thereof, taken to be the point $x = 0, y = 0$, it becomes $\log \sqrt{(x^2 + y^2)}$.

(2) At the boundary of the area $\xi = 0$.

(3) Throughout the area ξ satisfies the partial differential equation

$$\frac{d^2\xi}{dx^2} + \frac{d^2\xi}{dy^2} = 0.$$

In fact, if ξ be thus determined, it follows from (3) that we can on each differential element a , depending on $z = dx$, or determining η by the quadrature

$$\eta = \int \left(-\frac{d\xi}{dy} dx + \frac{d\xi}{dx} dy \right),$$

we have

$$\frac{d\eta}{dx} = -\frac{d\xi}{dy}, \quad \frac{d\eta}{dy} = \frac{d\xi}{dx},$$

and thence $\xi + i\eta$ is a function of $x + iy$.

Writing then $x_1 + iy_1 = e^{\xi + i\eta} = \phi(x + iy)$ suppose, we have an orthomorphosis of the area into the circle $x_1^2 + y_1^2 = 1 = 0$, viz. for the boundary of $z = 0$, and consequently $a + i\beta = e^{in}$, thence $a_n = \cos \eta, y = \sin \eta$, and therefore $a_n^2 + y_n^2 - 1 = 0$, or the point corresponds to a point on the circumference of the circle. Also, for a point z, y in the neighbourhood of $z = 0, y = 0$, we have

$$x_1 + iy_1 = e^{\log \sqrt{(x^2 + y^2)} + i\eta} = \sqrt{(x^2 + y^2)} e^{i\eta},$$

that is, $x = 0, y = 0$ corresponds to $x_1 = 0, y_1 = 0$ at the centre of the circle.

15. Reverting to the lemma, this in the first place asserts the existence of a function ξ satisfying the prescribed conditions, and next that the function is completely determinate, or say that there is but one function. As to the latter point, suppose that there are functions ξ, ξ' , satisfying the same conditions, then throughout the area, $\xi - \xi'$ is finite and continuous; by general principles in the theory of functions, this implies that the function has throughout the area a constant value, and since this value is for the boundary $\xi = 0$, the value is throughout the area $= 0$, that is, the function ξ is one and the same.

Some further considerations as to the continuity of η would be requisite in order to show that to the series of circles $a_n^2 + y_n^2 = a^2$, there correspond continuously increasing from 0 to 1, there correspond continuously enclosed curves surrounding the assumed origin and each other. Some further considerations as to the boundary of F , and thus to complete the proof that we thus obtain an orthomorphism of F into the circle, except only that to a point x_n, y_n on the circumference of F corresponds in general not a point on the contour of F , but a length thereof; but I abstain from a further discussion of the question.

Some valid considerations as to the continuity of η would be requisite in order to show that to the series of circles $x_n^2 + y_n^2 = a^2$, where a is continuously increases from 0 to 1, there correspond continuously enclosed curves surrounding the assumed origin and each other.

³This proof is taken from the paper, Christoffel, "Sul problema delle temperature stazionarie e la rappresentazione di una data superficie," *Annali di Matem.*, t. I. (1867), pp. 89—103.

In point, but I do not here a squarewise mesh sub of the area into the circle.

16. As a simple illustration, suppose that the area F is that of the circle $x^2 + y^2 = 1$, and that the excepted point is the point $x = 0, y = 0$, the centre of the circle. We have here a function ξ satisfying the conditions of the lemma, viz. this is $\xi = \log \sqrt{(x^2 + y^2)}$; the resulting value of η is $\eta = \tan^{-1} \frac{y}{x}$, and we have then $\xi + i\eta = \log(x + iy)$. Hence $e^{\xi + i\eta} = x + iy$, and the resulting orthomorphosis of the two circles is the identical one

$$x_1 + iy_1 = x + iy,$$

viz. we have

$$x_1 = x, \quad y_1 = y.$$

Writing then $x_1 + iy_1 = e^{\xi + i\eta} = \phi(x + iy)$ suppose, we have an orthomorphosis of the area into the circle $x_1^2 + y_1^2 - 1 = 0$, so the boundary $\xi = 0$, and consequently $x_1 + iy_1 = e^{i\eta}$. Thence $a_n = \cos \eta, y_n = \sin \eta$, and therefore $a_n^2 + y_n^2 = 1 = 0$, or the point corresponds to a point on the circumference of the circle. Also, for a point at $x = 0, y = 0$ in the neighbourhood of $x = 0, y = 0$, we have

$$x_1 + iy_1 = e^{in + i\eta(a + in) + i\eta} = \sqrt{(x^2 + y^2)} e^{i\eta},$$

that is, $x = 0, y = 0$ corresponds to $x_1 = 0, y_1 = 0$ at the centre of the circle.

So far considerations as to the continuity of η would be requisite in order to show that to the series of circles $x^2 + y^2 = e^{2\xi}$, where ξ is continuously increases from 0 to 1, there correspond continuously enclosed curves surrounding the assumed origin and each other, continuously diminishing area, and continuously approximating to a point; that the assumed-mesh number of meshes on the boundary of F , and thus to complete the proof that we thus obtain an orthomorphism of F into the circle, except only that to a point x_1, y_1 on the circumference of the circle there corresponds in general not a point of the contour F but a length of F . But I abstain from a further discussion of the question.

16. As a simple illustration, suppose that the area F is that of the circle $x^2 + y^2 = 1$, and that the excepted point is the point $x = 0, y = 0$, the centre of the circle. We have here a function ξ satisfying the conditions of the lemma, viz. this is $\xi = \log \sqrt{(x^2 + y^2)}$; the resulting value of η is $\eta = \tan^{-1} \frac{y}{x}$, and we have then $\xi + i\eta = \log(x + iy)$, and the resulting orthomorphosis of the two circles is the identical one

$$x_1 + iy_1 = x + iy,$$

viz. we have

$$x_1 = x, \quad y_1 = y.$$

17. The contraction of a circle to a given eccentric point has, in fact, been exhibited in the foregoing example in No. 5, but I assume the question from a different point of view. Writing for shortness a, a, a, a , for a point on the conjugate function, $z = a + iy, \bar{z} = z - iy$, and the conjugate function $f(\bar{z})$ which give for the conjugate value of $f(\bar{z})$, where also so general formula of (7) may be written in the form

$$x_1 + iy_1 = \frac{z - a}{1 - a\bar{z}},$$

viz. we have here a hyperbola, viz. its circumference $x^2 + y^2 - 1 = 0, x^2 + y^2 - 1 = 0$. In fact, repeating the verification, the change of sign of i gives

$$x_1 - iy_1 = \frac{\bar{z} - a}{1 - az}.$$

Here the product of the a -factors is

$$\frac{(z - a)(\bar{z} - a)}{1 - a(z + \bar{z}) + a^2},$$

which, if $x^2 = 1$, reduces to 1; the property exists, so do the a -factors, &c. multiplied for each. The correspondence which holds between a and θ above introduced will be apparent. For these circumstances, viz. on putting $a = z$, we have $a = z$, and there is no real point of any importance these are no real circumferences of the circles.

18. The most simple case is

$$x_1 + iy_1 = \frac{z - a}{1 - a\bar{z}}.$$

I assume here that a is real, an interior point of the circle $x^2 + y^2 - 1 = 0$ (viz. $x^2 - a^2 < 1$); hence $a = z$ gives a, a , viz. to the point $a = a$, $\nu = 0$, within the circle $x^2 + y^2 = 0$, there corresponds the centre of the circle $a_n^2 + y_n^2 = 1 = 0$, and we have in this equation the theory of the squarewise contraction of the entire circle $x^2 + y^2 = 1$ to the point $x = a$, $y = 0$ being the formula become

$$x_1 = \frac{x - a}{1 - ax}.$$

where $a < 1$, or if in the limit $a = 1$, may be taken to be positive, then $a < 1$.

19. To compare with a former investigation, writing $\xi + i\eta = \log(a_n + i\eta)$, we have

$$\xi + i\eta = \log \frac{a + i\beta a}{1 - i\beta} - \log \frac{\sqrt{((a - a) + j)}}{\sqrt{(1 - aay + aj)}} + i \tan^{-1} \frac{y}{a - a} - i \tan^{-1} \frac{ay}{1 - ax},$$

and consequently

$$\xi = \log \frac{\sqrt{(a - a)^2 + g^2}}{\sqrt{(1 - aay + aj)}}.$$

which is = 0 at the boundary $x^2 + y^2 = 1$, and has a finite and continuous throughout the area except in the neighbourhood of the point $x = a$, $y = 0$, for which it

= $\log \sqrt{(x^2 + y^2)}$; moreover, ξ satisfies the partial differential equation $\frac{d^2\xi}{dx^2} + \frac{d^2\xi}{dy^2} = 0$, and starting from the given value of ξ we should obtain the foregoing value of η , and thence $x_1 + iy_1 = e^{\xi + i\eta}$, that is, $x_1 + iy_1 = \frac{x + iy - a}{1 - a(x + iy)}$.

20. As further development of the solution, writing

$$x_1 - x_2 = \frac{x_1 - a - x(1 + a) + iy(1 - a)}{1 - a(z + iy)},$$

that is,

$$x_1 = \frac{x - a + iy(1 - a)}{1 - a^2(x^2 + y^2) - a^2 - a^2(x - a)}, \quad x_1 = \frac{x(1 - a^2)}{(1 - ax)^2 + a^2y^2},$$

and then assuming $x_1^2 + y_1^2 = \lambda^2$, we have for the contour corresponding therein

$$x^2 - a^2 + y^2 - \lambda^2((1 - ax)^2 + a^2y^2) = 0,$$

that is,

$$(x^2 + y^2)(1 - \lambda^2a^2) - 2x(a - \lambda^2a) + a^2 - \lambda^2 = 0,$$

or say

$$x^2 + y^2 - \frac{2a(1 - \lambda^2)}{1 - \lambda^2a^2}x + \frac{a^2 - \lambda^2}{1 - \lambda^2a^2} = 0,$$

viz. for further development of the solution, the circle $a^2 + y^2 - 1 = 0$ is the circle $x^2 + y^2 - 1 = 0$ corresponding to the eccentric point $x = a$, $y = 0$.

In further development of the solution, writing

$$\eta = \tan^{-1} \frac{y}{x - a} - \tan^{-1} \frac{ay}{1 - ax} = w,$$

that is,

$$\tan w = \frac{(1 - ax)y - ay(x - a)}{(1 - ax)(x - a) + ay^2},$$

showing that the contour curves all pass through two imaginary points

$$x = \frac{1}{a} \left(1 - \frac{1}{a}\right), \quad y = \pm \frac{i}{a} \left(\frac{1}{a} - a\right).$$

21. Writing next

$$x_1 = \frac{-x + a(1 + x^2) - a^2x + iy - iay - iay - iay}{(1 - ax)^2 + a^2y^2},$$

that is,

$$x_1 = \frac{-x + a(1 + x^2) - a^2x + iy - iay - iay - iay}{(1 - ax)^2 + a^2y^2}, \quad y_1 = \frac{y(1 - a^2)}{(1 - ax)^2 + a^2y^2},$$

then to the radius $x_1 - \theta y_1 = 0$ corresponds the radial

$$-x + a + ay - \theta y(1 - a^2) = 0,$$

that is,

$$(x - a) \left(a - \frac{1}{\theta}\right) + y^2 + \theta y \left(a - \frac{1}{\theta}\right) = 0,$$

which is a circle passing through the two real points $y = 0, x = a, x = \frac{1}{a}$, being the antipodes of the before-mentioned pair of imaginary points.

22. Hence for the contraction of the circle $x^2 + y^2 - 1 = 0$ to the interior point $x = a, y = 0$, calling this point A , and taking A' as image hereof (coordinates $x = \frac{1}{a}, y = 0$), the radials are circles through A and A' , but they cut the circle $x^2 + y^2 - 1 = 0$ at right angles; the contours are circles such that their tangents from each of the points A, A' are equal.

In other words, the distance AA' being hered (coordinates $x = \frac{1}{a}, y = 0$, and having thus for their common chord that line bisecting AA' at right angles, the equations of the contour circles and the radial circles assume the forms

$$x^2 + y^2 - 2hx + 1 = 0,$$

$$x^2 + y^2 - 2ky + 1 = 0,$$

respectively.

23. It is better, interchanging x, y and altering the constants, to write them in the form

$$x^2 + y^2 - 2hx + 1 = 0,$$

$$x^2 + y^2 - 2hy - 1 = 0,$$

viz. as no effect appearing in No. 5, these are the equations derived from $x + iy = \tan(u + iv)$ where $\tan u = a, \tan v = b$.

24. Passing now to the form

$$x_1 + iy_1 = \frac{z - a}{1 - az} \cdot \frac{z - \beta}{1 - \beta z},$$

to obtain the most simple instance, I write $a = 0, b = \sqrt{2}$, so that the form is

$$x_1 = \frac{i(x - \sqrt{2})}{1 - \sqrt{2}x},$$

⁴I made a transformation in figure 2, where the quasi-inversion is consequence of which after which a constant follows in y to A, B, D to A, B, D to A, B, D to A, B, D to A, B, D to A, B, D to A, B, D, &c.

or say

$$(x - \theta a)(x^2 + y^2) = (x + \theta a)a^2,$$

a circular cubic. In particular, for the line $x_1 = 0$, we have $\theta = 0$, and the curve is

$$x(x^2 + y^2 - a^2) = 0,$$

viz. this consists of the line $x = 0$, and of the circle $x^2 + y^2 - a^2 = 0$. We thus have for one of the radials, the line $x = 0$, that is, the chord which is real, the radials of figure 3.

The curve passes through the origin $x = 0, y = 0$, and the equation of the

tangent there is $x + \theta y = 0$. Moreover it passes through the centre of the circle, $x = \frac{1}{\sqrt{2}}, y = 0$, and the equation of the tangent there is easily found to be

$$x - \frac{1}{\sqrt{2}} - \theta y = 0,$$

the equation of the line being

$$\left(x - \frac{1}{\sqrt{2}}\right) - \theta y = 0,$$

We may find where the curve cuts the circle $\left(x + \frac{1}{\sqrt{2}}\right)^2 + y^2 - \frac{1}{2} = 0$, viz. this is $x^2 + y^2 - \sqrt{2}x = 0$, and substituting in the equation of the curve, we have

$$x(\sqrt{2}x - a) - \theta a(\sqrt{2}x + a)$$

or say

$$\left(x - \frac{1}{\sqrt{2}}\right)(x^2 + y^2) - \frac{1}{\sqrt{2}}\theta y(x^2 + y^2 - 2 \cdot \frac{1}{\sqrt{2}}x) = 0,$$

a hyperbola which for its intersections with the circle determines the points in question. There are two real intersections, but it is obvious that we are bounding the unshaded portion of the value of curves of which only a part is shown by quadrature of the intersection of the circle with the other branch of the cubic determines the contour cycle. There must thus be a circle (one of the contour cycles being a square) and shown in Plate XIII.

28. The formula

$$x_1 = \frac{x + a}{1 - ax}$$

has been considered for the case of an interior point a , the case of an exterior point a might be considered in like manner, but it is obvious that we pass from one to the other, by a transformation $\frac{1}{a}$ for a ; and it thus easily appears that, if a be an exterior point, viz. when $a^2 > 1$ that is, the exterior point, since this can be by the transformation $\frac{1}{a}$ to a , reduced to that of two interior points, but it is sufficient to consider this last case.

29. Considering then the form

$$x_1 = \frac{x + a}{1 - ax} \cdot \frac{x - \beta}{1 - \beta x},$$

where a, β are interior points, the most simple case is when $a = a, \beta = -a$, positive and < 1 . Here

$$x_1 + iy_1 = \frac{x - a^2 - a^2}{1 - a^2(a^2 - y^2)} + \frac{ay(1 - x^2 + y^2)}{(1 - x^2 - y^2)^2 + a^2(x^2 - y^2)^2 + 4a^2y^2},$$

and we have

$$x_1^2 + y_1^2 = \frac{(a^2 + y^2 - 2a^2 + a^2 - 4a^2y^2)^2}{1 - a^2(a^2 - y^2)^2 + 4a^2y^2}$$

$$= \frac{(x^2 + y^2 - a^2)^2 + 4a^2y^2}{1 - a^2(x^2 + y^2)^2 - 4a^2y^2},$$

which has for the circumference $x^2 + y^2 = 1$ the value $= 1$, the imaginary curve $x^2 + y^2 = 0$ corresponds to that of the unshaded curve. Writing now $a_n^2 + y_n^2 = w^2$ we have

$$(x^2 + y^2 - a^2)^2 + 4a^2y^2 = w^2\{(1 - a^2(x^2 + y^2)^2) + 4a^2y^2\},$$

a bicircular quartic which is of the figure, dividing, dissolving with a , and after some changes reducing itself to the form of a figure of eight in squarewise consider to the circle $a_n^2 + y_n^2 - a^2 = 0$. For $a = 1$, the curve breaks up into the two circles $a_n^2 + y_n^2 - 1 = 0$ and into the infinite curve $a^2 + y^2 = 1$, doubling the two circumferences

24. The Schwarzian orthomorphosis of the square into a circle is given (Memoir, p. 113) by the formula $X_1 + iY_1 = m(X + iY)$, where the modulus is $k = \frac{1}{\sqrt{2}}$, which corresponds to the angle 45° .

and for the circle, they are two sets of bicircular quartics, such that the portions within the circle have a configuration resembling that of the confocal ellipses and hyperbolas within the ellipse. To investigate the formula, it is convenient to introduce the function

$$X + iY = \sin^{-1}(s + ig),$$

we then have

$$x + iy = \sin(X + iY),$$

or say

$$x = \sin X \cosh Y, \quad y = \cos X \sinh Y,$$

so that to any given value of Y there corresponds the curve

$$\frac{x^2}{\cosh^2 Y} + \frac{y^2}{\sinh^2 Y} = 1,$$

and to

$$x = \sin X, \quad y = -\cos X, \quad \text{or,} \quad x_n = \sin X, \quad y_n = \cos X,$$

$$x = \sin X, \quad y = \cos X, \quad x_n = \cos X, \quad y_n = \sin X,$$

$$\theta = \sin X, \quad y = -\sin X(1 + iY),$$

then

$$x_1 = \sin X \cos iY, \quad y_1 = \cos X \sin iY,$$

giving

$$\frac{x^2}{\cos^2 X} - \frac{y^2}{\sinh^2 Y} = 1, \text{ if } 1 - x_n^2 = 0, \text{ that is, } X = (k + \frac{1}{2})\pi, \text{ or since}$$

$$\cos X = \sqrt{2} \cos X = \cos \cos X = \sqrt{2} \cos X$$

$$\sin Y = \sqrt{1 - \cos^2 X}, \quad \frac{dY}{dX} = \frac{cX}{\sqrt{1 - c^2 X^2}}, \text{ or } K = \frac{x^2 K}{a^2},$$

hence defining a as above, it appears that the elliptic polygons $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ corresponds to the circumference $a_n^2 + y_n^2 = 1$. The aside "Quartic and the graphical representation of the Elliptic Functions," Comb. Phil. Trans., t. XIV. (1889), pp. 105—120 (see p. 113) for the orthomorphosis of others into a circle.

25. The whole theory, and in particular Riemann's theorem before referred to,

are intimately connected with Cauchy's theorem, "If a function $f(z)$ is holomorphic over a simply connected plane area, and if z denote any point within the area, then

$$f(z) = \frac{1}{2\pi i} \int \frac{f(z)dz}{z - z'},$$

where z denotes $z = i\eta$, and the integral is taken in the positive sense along the boundary of the area." See Briot and Bouquet, *Théorie des fonctions elliptiques* (Paris, 1875), p. 136.

Here in order to obtain by means of the theorem the value of the function $f(z)$ for a given point of the boundary; viz. if z refers to a point P on the boundary, and z to a second point P_n in the boundary, but also the several points of $f(z)dz$, the form of this boundary, but also the several points P , throughout which correspond to the several points P of the first-mentioned boundary; or say we require to know the correspondence of the two boundaries: this being known, we have by the theorem the value of $f(z)$, that is, the point $a_n + i$, within the first area, which corresponds to the point z on the boundary, as a determinate value, determined by means of the given values of $f(z)$ along the boundary.

ON SOME PROBLEMS OF ORTHOMORPHOSIS

[921]

From Crelle's Journal der Mathem., t. cviii. (1891), pp. 263—277.

In the interesting Memoir, Schwarz, "Ueber einige Abbildungsaufgaben," *Crelle*, t. LXX. (1869), pp. 105—120, the author considers the orthomorphic transformation (or, as I call it, the orthomorphosis) of a square into the infinite half-plane, or into a circle, and of a rectangle into the same. It is of course easy to deduce the orthomorphosis of the rectangle into a circle; and thus, by giving a proper value to the modulus of the elliptic function involved in the formula, we obtain the orthomorphosis of the square into a circle, this solution (although equivalent thereto) being under a different form, and accompanied with a rectangle, and (although not necessary) considering for a moment the rectangle as a square, but to the circumference together with two two-cusped portions of a diameter. I propose to consider in I, these several transformations: II. relates to the orthomorphosis of a circle into a circle.

1. I. We are concerned with the elliptic functions for which $K' = 2K$, and also with the elliptic function sn of the lemniscatic form. The modulus in the former case is

$$k = \frac{\sqrt{2}-1}{\sqrt{2}+1} = (\sqrt{2}-1)^2; \quad k^2 = \sqrt{2}; \quad k^2 = -1 + 2\sqrt{2}; \quad \text{or say } k\sqrt{2} = \sqrt{2} + 1.$$

For the lemniscatic form, the modulus is $= \frac{1}{\sqrt{2}}$, and (with Gauss) we write $\frac{1}{2}\omega$ for the value of the complete function K or F_n ; thus we have $\frac{1}{2}\omega = K(\sqrt{2}-1)$.

$= K'\sqrt{2}$, where k has the foregoing special value: I notice the numerical values $\frac{1}{2}\omega = 1.311029$, $K = 3 + 2\sqrt{2} \cdot \frac{1}{2}\omega \cdot p = 1.682245$. The relation between the lemniscate function, and the sn with the foregoing value of k , is easily shown to be

$$\sqrt{2}snU = \frac{(c+1)sinl \cdot a\sqrt{2}}{(c-1)sinl \cdot a + \sqrt{2}} = 1.582245,$$

and it may be added that we have

$$snU - \frac{\sqrt{2}}{(c-1)sinl \cdot a + \sqrt{2}} = \frac{1}{2}sinl \cdot a(1 - cml \cdot a),$$

$$dnU = \frac{\sqrt{2}}{(c-1)sinl \cdot a + \sqrt{2}} \cdot \sqrt{1 + a \cdot (-cml \cdot a)} \cdot (1 + cml \cdot a),$$

2. The Schwarzian orthomorphosis of the rectangle into the infinite half-plane is given (Memoir, p. 113) by the formula $X_1 + iY_1 = \sin(X + iY)$, where the modulus is such that $K = 2K'$; viz. here—see figure 1 (X, Y) and figure 2 (x, y)—the rectangle $ABCD$, the half-diagonals whereof are each $= \frac{1}{2}\pi$, corresponds to the upper half-plane $a_n = 0$; in fact, $X_n = a$, $Y_n = 0$, where $\sin(X) = 0$, gives $x_n = \sin X$, $y_n = 0$. We have so that the entire side 4 of the rectangle $ABCD$ corresponds to the entire side D, C of the value of a , between the limits $a^2 = -1$, $a^2 = +1$, where (for the side A, B at this half-plane $y_n = 0$).

X_n has a and b symmetrically, in regard to the axis OY_n and in the case of the before-mentioned modulus $k = \sqrt{2} - 2 \cdot \sqrt{2}$ for which the rectangle $ABCD$ becomes a square. Fig. 1^a (X, Y_n).

square, the quasi-inversion may be so determined that the points A, B, C, D shall lie situate ortho (that is, in inclinations $\pm 45^\circ$, $\pm 135^\circ$) to the axis OY_n in the half-mentioned manner of figure 1, the four points A, B, C, D, A, AB of the centre of these circles, then forming A, A, B, C, D , as in the figure (it is for the figure of the lemniscate), it is at once obvious that the

boundary of the rectangle is a symmetrical one, will be situated, A and B symmetrically, and similarly for p' and b' . But we have

$$b = c \cdot \sqrt{2} \left(\frac{1}{a^2} - 1 \right) = \sqrt{2}(c - a)\sqrt{2}, \quad c = \sqrt{2}c + 1,$$

6. The general formula of the inversion whereby the rectangle $ABCD$ -axis is changed into a circumference is changed

$$X_1 + iY_1 = \frac{1 + i\sqrt{X(X_1 + iY_1)}}{M(X_1 + iY_1)} \cdot \frac{1 + i\sqrt{X(X_1 + iY_1)}}{M(X_1 + iY_1)} \cdot \frac{1 + i\sqrt{X(X_1 + iY_1)}}{M(X_1 + iY_1)} \cdot \frac{1 + i\sqrt{X(X_1 + iY_1)}}{M(X_1 + iY_1)},$$

where M is real. In fact, here this equation in the form

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$$X_1 + iY_1 = \frac{1 + i\sqrt{X(X_1 + iY_1)}}{M(X_1 + iY_1)},$$

we have

$$X_1 = iY_1 = \frac{1 - i\sqrt{X(X_1 - iY_1)}}{M(X_1 - iY_1)},$$

and thence $X_1^2 + Y_1^2 = 1 = 0$, that is, the infinite X_1 -axis is transformed into the circumference $X_1^2 + Y_1^2 = 0$.

Writing $Y_n = 0$, we have

$$X_n + iY_n = \frac{1 + ia(X_n + iY_n)}{M(X_n - iY_n)},$$

so that to a pair of points $(X_n, 0)$, there corresponds a pair of points (X_n, Y_n) symmetrically in regard to the axis OY_n .

The equation gives

$$M(X_n + iY_n) = (X_n + iY_n)(X_n - iY_n) - i(X_n + iY_n)$$

hence, writing this in the form

$$M(X_n + iY_n) = \frac{1 + i(X_n + iY_n)}{1 + i(X_n - iY_n)} \cdot \frac{(X_n + iY_n)}{1 + i(X_n + iY_n)},$$

we have

$$M(X_n - iY_n) = \frac{1 + i(X_n - iY_n)}{1 - i(X_n - iY_n)} \cdot \frac{(X_n - iY_n)}{1 - i(X_n - iY_n)},$$

and consequently $M^2(X_n^2 + Y_n^2) = 1$, if only $Y_n = 0$; that is, the circumference $X_n^2 + Y_n^2 = 0$ is transformed into the infinite X_n -axis.

Although for the purposes of the present memoir we require the coefficient M , yet there is no real loss of generality in assuming $M = 1$, and the transformation is then

$$X_1 + iY_1 = \frac{1 + i(X_n + iY_n)}{1 - i(X_n + iY_n)};$$

not post No. 12.

7. As already mentioned, the modulus (X_n, Y_n) and (X_n, Y_n) are connected by an equation of the foregoing form, and in the case of the square ($k = \sqrt{2}/2$) as before, the corresponding values for the points A, B, C, D may be tabulated as follows:

	$X_n + iY_n$	$X_n + iY_n$
$A,$	$X_n + iY_n = -$,	$X_n + iY_n = \frac{-1+i}{\sqrt{2}}$,
$B,$	$X_n + iY_n = -1$,	$X_n + iY_n = \frac{1-i}{\sqrt{2}}$,
$C,$	$X_n + iY_n = \frac{1}{\sqrt{2}}$,	$X_n + iY_n = \frac{1+i}{\sqrt{2}}$,
$D,$	$X_n + iY_n = -1$,	$X_n + iY_n = \frac{-1-i}{\sqrt{2}}$.

The proper value of M is $M = \sqrt{2} - 1$, viz. the required formula is

$$\sqrt{2} \cdot (X_1 + iY_1) = \frac{1 - i(X_n + iY_n)}{X_n + iY_n - 1}.$$

Thus for the point B , we should have $\sqrt{2}$, that is, $\sqrt{2} \cdot \frac{1-i}{\sqrt{2}+1} \cdot \sqrt{2-1} = \frac{(1-i)\sqrt{2}(\sqrt{2}-1)}{1-i\sqrt{2}} = \frac{\sqrt{2}-i}{1-i\sqrt{2}}$ which is right. And similarly for the point C , we should have

$$\frac{1}{\sqrt{2}} \cdot \frac{1-i(1+i)\sqrt{2}}{(1+i)\sqrt{2}-1} = \frac{1}{\sqrt{2}} \cdot \frac{1-i+1}{1-1+1+i\sqrt{2}} = \frac{2-i}{1+i\sqrt{2}}, \quad \frac{1+i}{\sqrt{2}},$$

which is right. And similarly for the points A and D .

8. We connect the square $ABCD$ of figure 1 with that of figure 4, by the equation

$$X_n + iY_n = \frac{1 + i(X_n + iY_n)}{X_n + iY_n - 1};$$

in fact, recollecting that $i(X_n + iY_n) = K_n$, $x_n = sn(a + iv)$, we have for the four points respectively

$$\begin{aligned} A, \quad & X_n + iY_n = K_n, \quad a + iv = K + iK_n, \\ B, \quad & X_n + iY_n = -K, \quad a + iv = -K + iK_n, \\ C, \quad & X_n + iY_n = K + iK_n, \quad a + iv = K, \\ D, \quad & X_n + iY_n = -K + iK_n, \quad a + iv = -K, \end{aligned}$$

values which satisfy the relation in question.

9. Hence writing

$$X + iY = U, \quad x + iy = u,$$

we have

$$U = \frac{1 + iu}{1 - iu},$$

which is the relation in No. 1 between the arguments U , u of the elliptic functions sn , cl . We have $X_n + iY_n = cl(u + iv)$, and consequently

$$\sqrt{2}(X_n + iY_n) = \frac{1 - i \cdot cl(u + iv)}{cl(u + iv) - 1},$$

Hence, substituting the value of $X_n + iY_n$, we find that, $X_n + iY_n$,

an equation which (multiplying on the left-hand side the numerator and the denominator each by $\frac{-1}{\sqrt{2}}$) may be written

$$\frac{1 - i \cdot cl(u + iv) + i \frac{-i}{\sqrt{2}}}{1 + i \cdot \frac{-i}{\sqrt{2}}(u + iv) - 1} = \frac{1 - i(X_n + iY_n)}{X_n + iY_n - 1},$$

that is, we have

$$X_n + iY_n = \frac{1 + i \cdot cl(u + iv)}{cl(u + iv) - 1},$$

an equation which shows that the figure A is in fact the figure 4 turned through an angle of 45° . We have thus proved the conclusion stated in No. 5 as to the connexion between the lemniscatic relation by the spaces equal in the relation, viz. from the rectangle.

10. It is convenient to collect here the several equations relating to the orthomorphosis of the square. We have

$$X_n + iY_n = sn(X + iY), \quad k = \sqrt{2} - 1;$$

$$a_n + (a_n = sn(a + iv);$$

$$\sqrt{2}(X_n + iY_n) = \frac{1 - i(X_n + iY_n)}{X_n + iY_n - 1};$$

$$\sqrt{2} \cdot a_n = \frac{-1 - ia_n}{X_n + iY_n - 1};$$

$$X_n + iY_n = \frac{1 + ia_n}{1 - ia_n};$$

$$X_n + iY_n = \frac{1 + i(X_n + iY_n)}{1 - i(X_n + iY_n)},$$

which are the equations connecting together the coordinates of the four figures.

11. I examine more in detail the above-mentioned transformation

$$X_n + iY_n = \frac{1 + i(X_n + iY_n)}{X_n + iY_n - 1},$$

or, the foregoing figures 1 (X) and 2 (X, Y, n), in which we may regard the two circles as having each of them the same radius unity, changing the sign of i , the equation gives

$$X_n - iY_n = \frac{1 - i(X_n - iY_n)}{X_n - iY_n - 1},$$

and we hence find

$$X + Y = \frac{X_n^2 + Y_n^2 + 2X_n - 1}{X_n^2 + Y_n^2 - 2X_n + 1},$$

consequently if $X^2 + Y^2 = 1$, then also $X_n^2 + Y_n^2 = 0$; or the transformation changes the form of these imaginary circles into the second of them: that is, the orthomorphosis of the circle $X_n^2 + Y_n^2 = 0$ into the concentric orthomorphosis of the circle $X_n^2 + Y_n^2 = 1$. Or we have

$$X_n^2 + Y_n^2 - 2X_n + 1 = 0, \quad X_n^2 + Y_n^2 + 2X_n + 1 = 0,$$

values which give the foregoing expression for $X^2 + Y^2$. We can no further obtain

$$X^2 + Y^2 - 1 = \frac{2\sqrt{2}}{X_n^2 + Y_n^2 + 2X_n + 1}, \quad Y_n = \frac{X_n^2 + Y_n^2 + 2X_n + 1}{X_n^2 + Y_n^2 + 2X_n + 1} = 0;$$

and it then appears that the circumferences $X_n^2 + Y_n^2 = 0$ and $X_n^2 + Y_n^2 - 1 = 0$ correspond to each other.

correspond to each other. These are circles passing through the pairs of points ($Y_n = 0, X_n = \pm 1$), ($Y_n = 0, X_n = \pm 1$) respectively; or imagining them in the same figure, say they are circles which belong each to the centre of the figure, with the line $X_n = 0$, and which moreover cut at right angles the circles $X_n^2 + Y_n^2 + 2X_n + 1 = 0$, $X_n^2 + Y_n^2 + 2X_n + 1 = 0$. But attending more carefully to the nature of the correspondence, it is to be observed that taking X positive, and not at the value of $a_n = \pm 1$, then the corresponding more on the upper half-circle from $a_n = \pm 1$ to $a_n + 1$ on the upper semicircle LFM ; and that, corresponding hereto in figure 2, as are LJM lying wholly within the lower semicircle LDM ; and, that, corresponding herein in figure 2, as are LFM on LM to the diameter LM of the first circles LMF ; that is, the upper

half-plane bounded below by the upper semicircle LFM corresponds to the lower semicircle LDM in the first figure; that is, the upper half-plane bounded below by the lower semicircle LDM . Thus the upper semicircle LFM of figure 2 corresponds to the lower semicircle LDM of figure 1; and as for the lower semicircle LDM of figure 3, the upper half-plane bounded below by the lower semicircle LDM in figure 2, and below by the lower semicircle LDM in figure 2, are belonging wholly inside the upper semicircle LJM ; that is, in figure 2, are by figure 3, the upper semicircle LFM of figure 2; corresponding to the semicircle LJM of figure 3 corresponding to the semicircle LDM of figure 1, viz. the lower semicircle LDM of figure 1 corresponds to the upper semicircle LDM of figure 3; the upper half-plane bounded by the lower semicircle LDM in figure 2 corresponds to the lower part of the upper half-plane bounded above by the lower semicircle LDM in figure 3; the upper half-plane $a_n = 0$ here corresponds to the upper half-plane of the upper half-plane bounded by the upper semicircle LJM in figure 3.

12. In explanation of No. 4, observe that, starting from the equation

$$X_n + iY_n = m(X + iY), \quad k = \sqrt{2} + 1,$$

and writing $mX = P$, $miY = iQ$ (the analytical formula obtains for any real value of k), and assuming a system of values $X^2 = 1$ and $Y^2 = 1$ each passing through the points $X = 0, m = 1$, and these of the other set through the antipodes $X_n = \pm 1, Y_n = 0$.

This being so, the equation of the two circles is

$$a^2 + y^2 - 2p, \cos \mu, \quad p_1 = 1,$$

$$X_n + iY_n = \frac{P\sqrt{1 + Q^2 \cdot 1 + k^2 Q^2} + iQ\sqrt{1 + P^2 \cdot 1 - k^2 P^2}}{1 + k^2 P^2 Q^2},$$

that is,

$$X_n = \frac{P\sqrt{1 + Q^2 \cdot 1 + k^2 Q^2}}{1 + k^2 P^2 Q^2}, \quad Y_n = \frac{Q\sqrt{1 + P^2 \cdot 1 - k^2 P^2}}{1 + k^2 P^2 Q^2}.$$

and thence

$$X_n + Y_n = \frac{P + iQ}{1 + k^2 P^2 Q^2} \cdot (X_n + Y_n \text{ be put } = mP);$$

hence

$$P^2(1 - k^2 P^2)m^2 = -Q^2 \cdot Q^2(1 - k^2 P^2)m^2 = -P.$$

Now considering in figure 1 say line in the rectangle parallel to the axis of X , that is, taking Y constant and therefore also Q constant, and proceeding to eliminate P , we have

$$1 - P^2 = \frac{1 + k^2 Q^2(1 - m^2)}{1 - k^2 m^2 Q^2}, \quad 1 - k^2 P^2 = \frac{1 + k^2 Q^2(1 - m^2)}{1 - k^2 m^2 Q^2}, \quad 1 + k^2 P^2 Q^2 = \frac{1 + k^2 Q^2}{1 - k^2 m^2 Q^2},$$

and consequently

$$X_n = \sqrt{1 + Q^2 \cdot 1 + k^2 Q^2} \cdot \frac{1 + Q^2(1 - k^2 m^2 Q^2)}{1 + k^2 Q^2},$$

$$Y_n = \frac{Q\sqrt{1 + Q^2(1 - k^2 m^2)}\sqrt{1 + Q^2 - 2kY(1 + Q^2)}}{1 + k^2 Q^2},$$

giving $Y_n^2 + 1 = \frac{1}{m^2}$ of these in terms of Q and m , $X_n^2 + Y_n^2$. The former of these equations, replacing thereby η^2 by its value, gives

$$(X_n^2 + Y_n^2 - 2kX_n - 2kY_n) + \frac{Q^2}{m^2} = 0,$$

viz. we have thus the equation of the curve, a bicircular quartic which in figure 2 corresponds to the line parallel to the axis of X in figure 1.

In particular, for the line LM of figure 1, we have $Y = \frac{1}{k}$ and thence $Q = \frac{1}{km}$; that is, $Q = \frac{1}{k}$, and hence we have $\Delta = \frac{1}{k}$: the equation of the bicircular quartic is

$$(X_n^2 + Y_n^2)^2 = \frac{2}{k}(X_n + Y_n) + \frac{1}{m^2} = 0;$$

viz. this is the circle $X_n^2 + Y_n^2 = \frac{1}{k}$ twice repeated, and we have then this circle, or rather the half-circumference LFM of figure 1, corresponding to the line LM of figure 1. More precisely, to the line CD of figure 4, there corresponds the quarter-circumference CD

and thence

$$X_n + Y_n = \frac{(1 + iP)\sqrt{P^2 - 1} - (1 + iP)\sqrt{1 + k^2 P^2}}{4(1 + k^2 P^2)(1 + iQ)},$$

that is, $X_n + Y_n = 0$ as before. It is easy to see that the points A, O, B of figure 1 correspond to the points A, O, B of figure 2, and hence that the area of the rectangle $AOBF EA$ of figure 1 corresponds to that of the semicircle $AOBF EA$ of figure 2.

Returning to the question $X_n + iY_n = m(X + iY)$, if we write therein successively $m = \alpha$ and $m = \beta$, we obtain two values of the rectangle $AOBF EA$ in figure 2, parallel to AB , equal in their successive positions (α, β) , which we have in their successive horizontal positions inside the rectangle. We have moreover, in like manner, similarly for the parallel sides AD and BC as cause of which, viz., similar curves are the parallel sides of the rectangle. *The points A, O, B correspond to the cusps A, O, B of the same curve*, viz. of figure 2, that is, the cusps in the inner branches of the curve corresponding to $AOBF EA$ in figure 1, and so on.

13. Similarly in explanation of No. 5, observe that, starting from the equation

$$x_1 + iy_1 = ml(x + iy) \text{ and writing } onl = x, \ ml = y \text{ and } y = my, \text{ we have}$$

$$x_1 + iy_1 = \frac{P\sqrt{1 - Q^2}\sqrt{1 + Q^2} + iQ\sqrt{1 + P^2}\sqrt{1 - P^2}}{1 + P^2Q^2},$$

that is,

$$x_1 + iy_1 = \frac{P\sqrt{1 - Q^2}\sqrt{1 + Q^2}}{1 + P^2Q^2}, \quad y_1 = \frac{Q\sqrt{1 + P^2}\sqrt{1 - P^2}}{1 + P^2Q^2},$$

and thence

$$x_1^2 + y_1^2 = \frac{P^2 + Q^2}{1 + P^2Q^2}, \quad x_1^2 - y_1^2 = \frac{P^2 - Q^2 - P^2Q^4 + Q^2P^4}{(1 + P^2Q^2)^2},$$

writing $x + iy = \frac{1}{2}\omega$, we have

$$sn \frac{1}{2}\omega = \frac{P\sqrt{1 - Q^2}}{\sqrt{1 + P^2Q^2}}, \quad cn \frac{1}{2}\omega = \frac{\sqrt{1 - P^2}}{\sqrt{1 - Q^2}}, \quad dn \frac{1}{2}\omega = \frac{\sqrt{1 + P^2 - 1}}{\sqrt{1 + Q^2}}, \quad dn \frac{1}{2}\omega = \sqrt{1 + Q^2},$$

the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is transformed into the circle $x_1^2 + y_1^2 = 1$. More precisely, to the line CD of figure 4, there corresponds the quarter-circumference CD